Academic Year 2026 – 2027
Course: Creativity & Innovation - Spring 2026
Student: Muhammad Ibrahim

LinkConnect: A Modular Open-Source Cable Management System

Muhammad Ibrahim

7th May 2026

Abstract

Cable clutter in modern workspaces is a widespread problem that existing solutions – cable trays, Velcro sleeves, and zip ties – address only superficially, grouping cables together without separating them, restricting airflow, and offering no path for repair or upgrade. This report presents **LinkConnect**, a modular, parametric cable management system that routes each cable through its own discrete channel, developed in SolidWorks as a seven-part parametric assembly, validated through an 80-step finite element load sweep and a lumped-capacitance thermal simulation, and released as open-source hardware on GrabCAD. Structurally, the ABS variant remains safe up to $\lambda \approx 0.70$ before the pin yields by design as a sacrificial, user-replaceable component, while the Aluminium variant remains within safe limits across the full load range. Thermally, 493 perforations on the plate bottom increase the effective cooling surface by 129%, reducing steady-state temperature from 95.5 °C to 59.5 °C under a 100 W load, keeping the ABS plate within its safe operating range. A single tile can be fabricated at home for under AED 37, or produced at scale for approximately AED 17, with the open-source release enabling decentralised fabrication and a right-to-repair model where any component can be individually replaced. All six project objectives were completed in full.

Keywords: cable management, modular design, additive manufacturing, FEA, open-source hardware

Contents

1 Introduction

Modern workspaces and home offices host a growing number of electronic devices: laptops, monitors, docking stations, chargers, and networking equipment. Each device adds cables, and the resulting clusters overlap, twist, and strain connection points. Unmanaged cables lead to connector fatigue, poor airflow, insulation wear, and general clutter that reduces both safety and efficiency.

Existing solutions only partially help. Under-desk cable trays collect cables into a single channel but do not separate them, so cables still compress against each other. Velcro sleeves bundle cables together, restricting heat dissipation and making individual cable removal difficult. Neither solution addresses the underlying problem of cable interaction and load distribution.

LinkConnect is a modular, parametric cable management system that routes each cable along discrete channels, held in place by clips that can be freely positioned and locked anywhere on the tile grid. The system is designed to be upgraded, repaired, and extended at home. It was developed in SolidWorks, validated through structural and thermal simulation, and released as open-source for distributed fabrication on consumer 3D printers. This report covers the design, analysis, and distribution of the system.

2 Project Goals & Objectives

The goal of LinkConnect is to build a modular cable management system that is structurally sound, easy to fabricate at home, repairable, and openly distributed. The following objectives were set at the start of the semester:

1. **Develop a fully parametric CAD model** in SolidWorks, covering all parts and the top-level assembly, adjustable for different cable diameters and densities.
2. **Perform structural analysis** via load estimation and finite element analysis to validate the design under realistic cable loads, identify points of failure, and ensure that individual parts can be replaced at home when something breaks. This also supports right-to-repair principles, making upgrades and repairs straightforward for the end user.
3. **Select appropriate materials** based on cost, availability, environmental impact, and structural performance.
4. **Design discrete cable pathways** that physically separate cables and prevent entanglement.
5. **Release the design as open-source** through GrabCAD to enable decentralised fabrication.
6. **Establish a community feedback loop** to capture suggestions for improvement and guide future revisions.

3 Methodology & Approach

The project was carried out in three phases.

Phase 1: Design & Modeling (Weeks 6–8). A parametric CAD model was built in SolidWorks from the bottom up. Each part was modelled individually with dimensions tied to sketch parameters, then assembled into a top-level assembly. This allowed rapid iteration without rebuilding geometry from scratch.

Phase 2: Engineering Analysis & Validation (Weeks 9–12). Structural load estimation and finite element analysis were performed to validate the clip-pin interface under realistic cable loads. A parametric load sweep compared ABS and Aluminium across 80 load steps. A separate thermal simulation evaluated steady-state temperatures and heat dissipation for both materials in perforated and solid configurations.

Phase 3: Distribution & Community Engagement (Weeks 13–14). The complete design package was uploaded to GrabCAD with full documentation. A community feedback channel was opened and early responses were reviewed to identify improvements for future iterations.

4 Results Summary

This section summarises the key results from each project phase. Detailed findings are presented in Sections 5 through 7.

4.1 Design & Modeling Accomplishments

A complete parametric CAD assembly was developed in SolidWorks, comprising seven interacting parts: Plate (18 cm × 30 cm tile), Clip (three-hole cable carrier), backclip (rear support), Rail_A and Rail_B (interlocking connectors), Pin (load-bearing connector), and SeparatingRod (lateral spacing element). The modular design enables seamless expansion and reconfiguration without fasteners or adhesive. A single plate unit measures 18 cm × 30 cm, with the overall footprint not exceeding 20 cm × 30 cm once rails and connectors are accounted for. Users deploying multiple tiles can expand the system indefinitely in both horizontal directions, so no upper-bound area estimate is given. All detailed part dimensions are provided in the engineering drawings in the Appendix.

4.2 Engineering Analysis & Validation Accomplishments

Structural validation via load estimation and FEA demonstrated:

- ABS variant yields at $\lambda \approx 0.70$; Aluminium variant remains safe across the full design load range ($\lambda = 0$ to 1.0).
- Peak pin reactions at full load reach 2116.4 N.
- Maximum tip deflection is 2.5 mm, acceptable for under-desk mounting.

The FEA assumes **100% infill density** throughout all parts. At home, this is achievable with any consumer-grade FDM printer using ABS, PLA, or any high-rigidity thermoplastic. 100% infill is strongly recommended to match the structural behaviour validated in the simulation; lower infill densities reduce the load-bearing capacity below the validated values.

4.3 Material Selection Accomplishments

4.3.1 Home Fabrication Route

For home printing, **PLA** is recommended due to its safety profile. Unlike ABS, PLA produces negligible harmful fumes and is derived from renewable plant-based sources, making it suitable for enclosed home environments without specialist ventilation. ABS produces styrene fumes during printing, classified as Group 2B by the International Agency for Research on Cancer (IARC) at sustained concentrations, and should only be used in a well-ventilated space or an enclosed printer with HEPA/activated-carbon filtration. ABS remains the material of choice for the production variant due to its superior mechanical properties and heat resistance. Home users are therefore recommended to use PLA as a safe substitute, accepting a marginal reduction in structural performance.

4.3.2 Mass Manufacturing Route

For mass manufacturing, the material allocation is as follows:

- **Plate, Clip, backclip, Pin** – ABS plastic, injection moulded or high-output FDM printed.

— **Rail_A and Rail_B** – Aluminium, die-cast with post-process finishing (anodising, powder coating).
— **SeparatingRod** – Cast iron, providing high rigidity and dimensional stability at low cost.

Two material routes were structurally validated:

— **ABS plastic** ($\approx$ 44 MPa yield): suitable for home FDM fabrication; requires lower per-module cable counts as a safety factor at reduced infill.
— **Aluminium 6061** ($\approx$ 70 MPa yield): superior strength and thermal properties; used for rails and suitable for industrial deployments.

4.3.3 Environmental Impact Assessment

— **PLA**: Biodegradable under industrial composting conditions, derived from renewable feedstocks. Minimal toxicity during printing. Best choice for home users from an environmental standpoint.
— **ABS**: Petroleum-derived, not biodegradable. Produces styrene vapours during FDM printing (IARC Group 2B at sustained exposure). Requires enclosed printing or good ventilation at home. Recyclable through standard plastic streams (code 9). Its durability reduces replacement frequency, partially offsetting its embodied carbon.
— **Aluminium 6061**: High embodied energy in primary production. However, recycled aluminium requires only $\approx$ 5% of the energy of primary production, and recycling infrastructure is widely available. Non-toxic, corrosion-resistant, and infinitely recyclable.
— **Cast iron (SeparatingRod)**: High embodied energy but extremely durable and fully recyclable. No toxicity concerns during use.

A key environmental advantage of LinkConnect is its **decentralised manufacturing** philosophy. By releasing the design on GrabCAD, users fabricate, repair, or upgrade individual components without discarding the whole assembly. If a single clip breaks, the user prints only that clip. This right-to-repair approach also eliminates the carbon footprint of global shipping and centralised warehousing.

4.4 Cost Analysis

4.4.1 Home Fabrication Cost Estimate

The following estimates assume a single tile module printed at 100% infill. Filament costs are based on UAE retail pricing of AED 85–110 per 1 kg spool (approximately AED 0.09 per gram). The UAE DEWA/FEWA residential electricity tariff of AED 0.23 per kWh is applied, with a consumer FDM printer drawing approximately 200 W during printing. Printer purchase cost is not included. The user requires access to a printer with a build plate of at least 20 cm $\times$ 30 cm.

Key observations:

— A single tile module costs under **AED 37** in materials and electricity.
— Filament cost dominates; electricity at UAE tariffs is negligible at under AED 0.50 per module.
— A single 1 kg spool yields approximately 2–3 complete tile modules at 100% infill.

4.4.2 Mass Manufacturing Cost Estimate

At production scale, costs reduce significantly through high-output injection moulding for ABS parts, die-casting for Aluminium rails, and sand casting for Cast Iron SeparatingRods. Post-processing (deburring, anodising, surface smoothing) adds a finishing cost per unit.

Key observations:

— Mass manufacturing reduces the per-unit cost to approximately **AED 17**, roughly half the home fabrication cost.

Table 1 – Home fabrication cost per complete tile module at 100% infill, UAE pricing in AED.

Component	Est. Mass (g)	Material cost (AED)	Print time (hrs)
Plate	180	16.20	4.00
Clip × 8	96	8.64	2.00
backclip × 2	20	1.80	0.50
Pin × 8	16	1.44	0.50
SeparatingRod × 4	24	2.16	0.50
Rail_A	30	2.70	0.75
Rail_B	30	2.70	0.75
Subtotal (material)	**396**	**35.64**	**9.00**
Electricity (9.0 hrs × 0.2 kW × AED 0.23/kWh)	–	0.41	–
Grand Total	**396 g**	**AED 36.05**	**9.0 hrs**

Table 2 – Mass manufacturing cost breakdown per tile at production scale (>1000 units), in AED.

Component	Process	Material	Est. unit cost (AED)
Plate	Injection moulding	ABS	4.40
Clip × 8	Injection moulding	ABS	2.95
backclip × 2	Injection moulding	ABS	0.55
Pin × 8	Injection moulding	ABS	0.37
SeparatingRod × 4	Sand casting	Cast Iron	1.47
Rail_A	Die casting + anodising	Al 6061	2.20
Rail_B	Die casting + anodising	Al 6061	2.20
Assembly & QC	–	–	1.84
Finishing / packaging	–	–	1.10
Total unit cost			**AED 17.08**

- The cost advantage of home fabrication lies in **accessibility**: no minimum order quantity, no shipping, and immediate availability.
- The GrabCAD open-source release means users with printer access can fabricate the entire system at no product cost, reinforcing the right-to-repair and decentralised manufacturing ethos of the project.

4.5 Distribution & Community Engagement Accomplishments

The design package was successfully released on GrabCAD with:

- Complete SolidWorks assembly and all seven parts in `.SLDPRT` format.
- Documentation covering design intent, recommended print parameters, and assembly instructions.
- An embedded 3D viewer for web-based inspection.
- An open feedback channel; the first community comment was received within one week of release.

5 Design & Modeling

5.1 CAD Development

5.1.1 Plate

The Plate is the primary structural tile of the system, measuring 300 mm × 180 mm × 25 mm. Its top surface features a dense waffle-pattern grid with 4 mm pin pockets, into which clip-pin assemblies are inserted and slid freely in the x-y plane. The long edges carry interlocking slots for the rail components, and the underside is heavily perforated for thermal management and weight reduction.

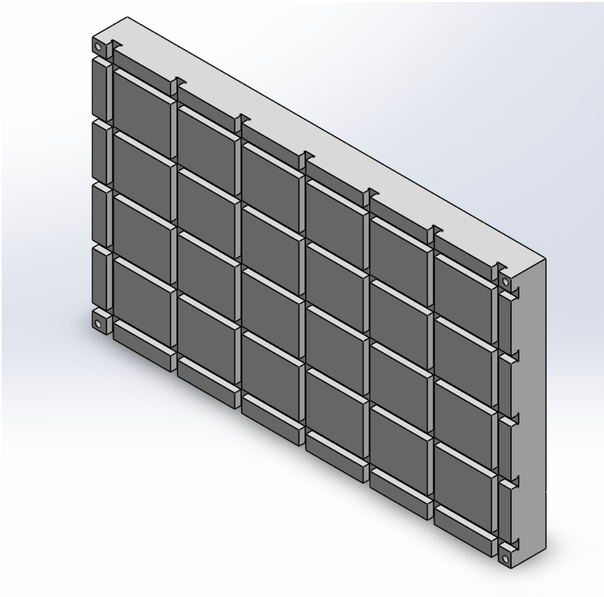

Figure 1 – Engineering drawing of the Plate. Top view (left) shows the waffle-pattern grid across the full 300 mm × 180 mm surface. The side view (centre) and isometric (right) show the 25 mm thickness and interlocking edge profile.

5.1.2 Clip

The Clip is the cable-carrying element, measuring 36.6 mm × 28 mm × 22.85 mm with a rounded top profile (R6 mm). It carries three through-holes of diameters 13 mm, 11 mm, and 7 mm, accommodating a range of cable sizes from large power leads to slim data cables. A row of four 3.5 mm mounting holes on the rear face accepts the Pin component, securing the clip into the plate grid.

5.1.3 Backclip

The backclip is a small oval bracket measuring 8 mm × 8 mm with a 4 mm corner radius and a 20 mm overall height. It carries two 3.5 mm mounting posts on its rear face and is used to connect up to four plates together at their corners, forming a larger multi-tile assembly without fasteners. It also acts as a cover for the clip mounting interface, improving the finished appearance of the assembly.

5.1.4 Pin

The Pin is the load-bearing fastener that secures each clip into the plate grid. It has a 9.5 mm top cap, 2.5 mm flange height, a 6.5 mm stem of 4.5 mm length, and a 7 mm base. The pin is inserted through the bottom of the clip and into the plate pocket, locking the clip in its chosen x-y position. It is the

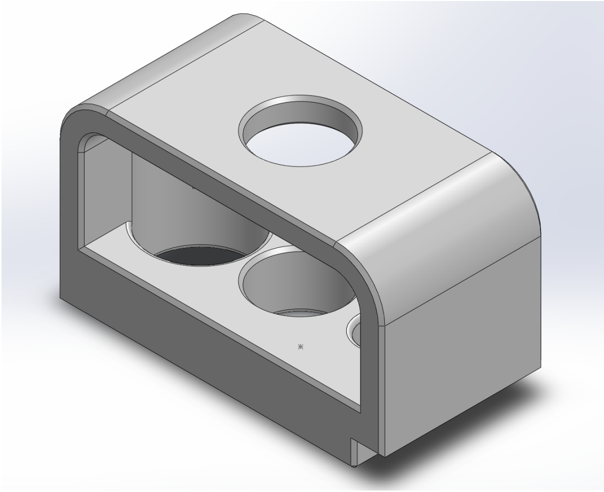

Figure 2 – Engineering drawing of the Clip. The front view (top left) shows the three cable holes (13 mm, 11 mm, 7 mm) and the 36.6 mm × 28 mm body. The isometric (bottom right) shows the mounting interface on the rear face.

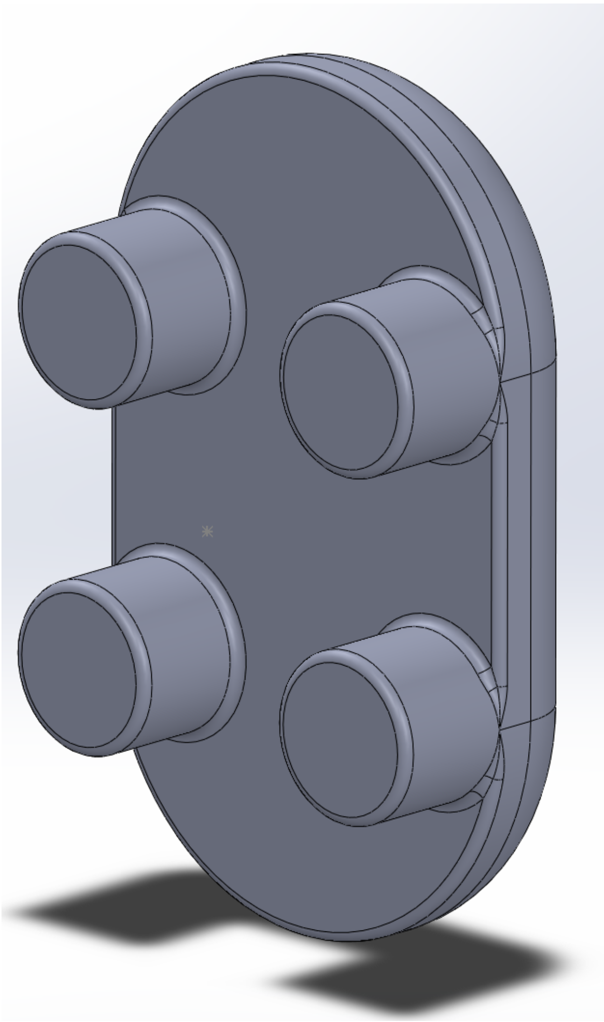

Figure 3 – Engineering drawing of the backclip. The top view (left) shows the 8 mm × 8 mm oval profile with R4 mm. The side view (top right) shows the two 3.5 mm mounting posts at 20 mm spacing. The isometric (centre) shows the assembled form.

intentionally sacrificial component: under overload it yields first and can be individually replaced at home.

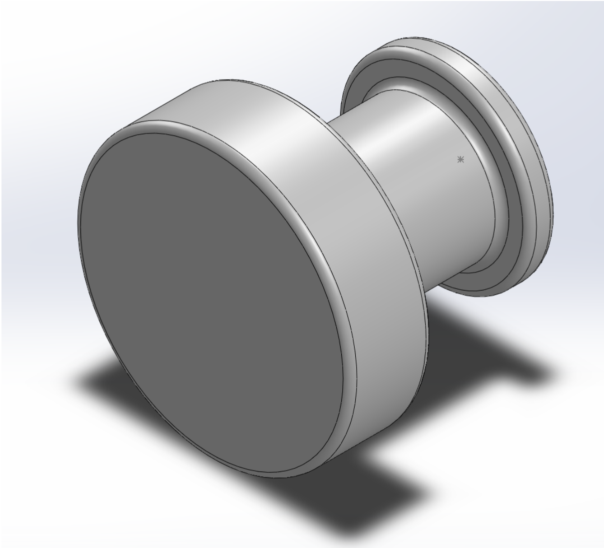

Figure 4 – Engineering drawing of the Pin. Top view (top left) shows the 9.5 mm cap and 6.5 mm stem. Bottom view (bottom left) shows the 10 mm outer, 7 mm inner, and 9.5 mm mid profile. The isometric (bottom right) shows the full assembly form.

5.1.5 Rail_A and Rail_B

Rail_A is a flat longitudinal bar measuring 300 mm × 28 mm × 3.5 mm, with a 60 mm mounting tab carrying four 6 mm holes. Rail_B is a vertical bracket, 180 mm tall, 28 mm wide, and 38 mm deep with a 3 mm wall and a 60 mm mounting plate. Rail_A attaches along the long edge of the plate and Rail_B at the short end; together they lock all clip pins in the grid channel against lateral movement, completing the structural frame of the tile.

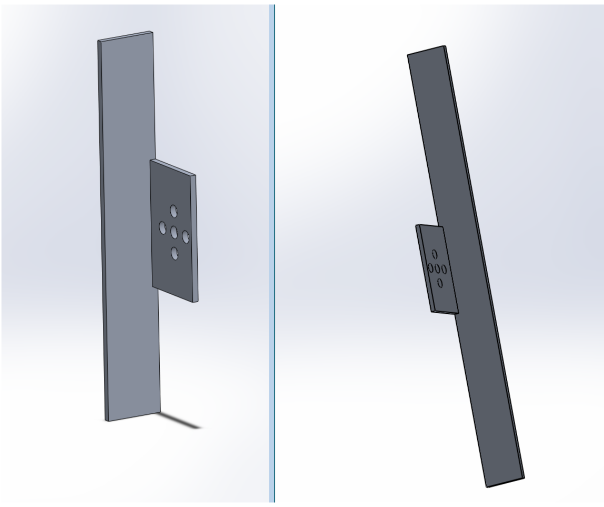

Figure 5 – Engineering drawings of Rail_A (left) and Rail_B (right). Rail_A spans the full 300 mm plate length; Rail_B provides the 180 mm end constraint. Both carry 6 mm mounting holes on a 60 mm pitch.

5.1.6 SeparatingRod

The SeparatingRod is a simple 4 mm cylindrical rod, 100 mm long. It is inserted between adjacent plates to maintain a fixed vertical gap when tiles are stacked on top of each other, enabling multi-level cable routing configurations. It is optional in single-layer installations but essential for stacked deployments.

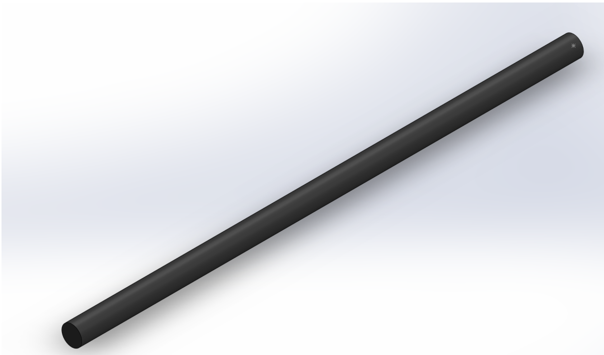

Figure 6 – Engineering drawing of the SeparatingRod: 4 mm × 100 mm cylindrical rod used as a vertical spacer between stacked tile layers.

5.1.7 Full Assembly

A complete LinkConnect assembly is built as follows. One or more plates are laid out and joined at their corners using backclips, forming a continuous tiled surface. For each cable position, a Pin is inserted into the bottom of a Clip, and the pin-clip unit is dropped into the waffle grid of the plate, where it can slide freely in the x-y plane to any desired position. Once all clips are positioned, Rail_A is attached along each long edge and Rail_B at each short end, locking all pins in the grid channels and completing the structural frame. If a multi-level configuration is required, SeparatingRods are inserted between tile layers to maintain the vertical gap. Figure 7 shows the complete assembled system.

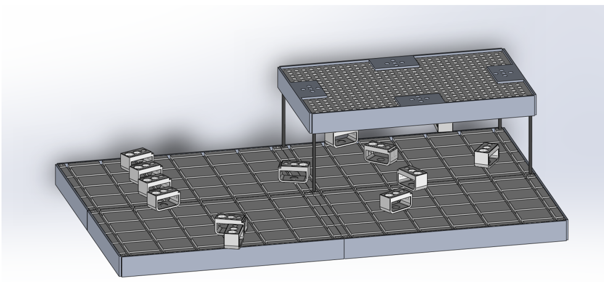

Figure 7 – Complete LinkConnect assembly showing two tiles joined by backclips, with multiple clip-pin units positioned in the waffle grid, Rail_A and Rail_B securing the perimeter, and the perforated cover plate on top.

5.2 Modular Interlocking System

Each tile measures 18 cm × 30 cm and is designed to connect with adjacent tiles via backclips at the corners and interlocking rail slots along the edges. Because all tiles are geometrically identical, the system can be extended in any direction without additional hardware, making it straightforward to fabricate in batches and for users to expand as their workstation grows. The clip-pin units can be repositioned at any time by releasing the rails, sliding the clips to new grid positions, and re-securing the rails.

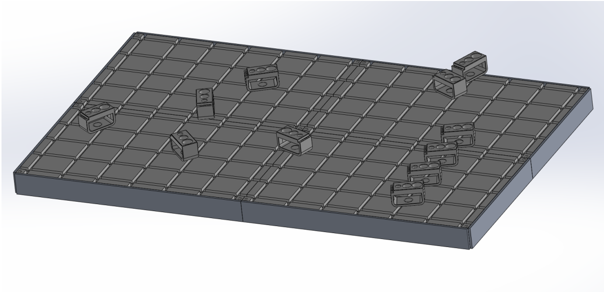

Figure 8 – Multi-tile modular assembly. Backclips join plates at the corners, rails lock the perimeter, and clip-pin units are positioned freely within the waffle grid. The system extends indefinitely in both horizontal directions.

5.3 Discrete Pathing

Rather than collecting cables into a single tray, LinkConnect gives each cable its own discrete channel through one of the three holes in the Clip. The three hole sizes (13 mm, 11 mm, 7 mm) cover the full range from large power leads to slim data cables, and the triangular arrangement maximises separation while minimising the clip footprint. Because each clip is independently positioned in the grid, the user can route any cable to any location on the tile surface, and individual cables can be added or removed without disturbing any neighbouring cables.

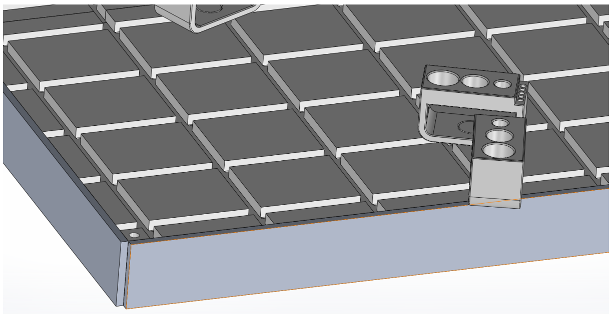

Figure 9 – Discrete pathing: three-hole clip (left), cross-sectional view showing the three cable diameter channels (centre), and a populated tile with clips at user-selected grid positions (right).

6 Engineering Analysis & Validation

6.1 Structural Load Estimation

The system was modelled as a structural component subjected to distributed loading from suspended cables. Typical workstation cables have a mass of roughly 0.05 to 0.15 kg/m, and a single module supports between eight and fifteen cables. The combined cable weight, with a conservative dynamic factor applied for handling, gives the total distributed load on the system.

Unlike conventional cable trays, LinkConnect uses a **waffle-pattern grid** on the top surface of each tile. Clip components insert their mounting pins into this grid and can slide freely in both lateral directions. The Rail components lock the clips in their chosen positions along the tile edges, giving the user complete freedom to configure the layout. Figure 10 shows a two-tile assembly with clips at various grid positions, and Figure 11 shows the system in a vertical wall-mount configuration.

A parametric load sweep was performed on the clip-pin contact interface, increasing the load factor λ from 0 to 1 across 80 steps, corresponding to a maximum prescribed displacement of 2.5 mm. The ABS

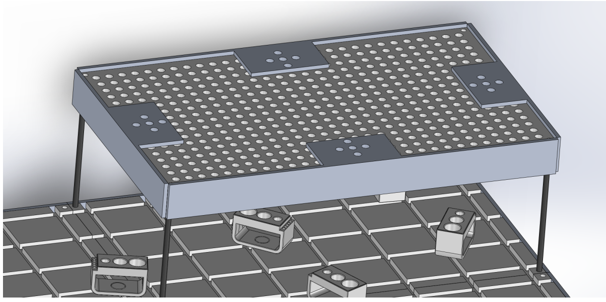

Figure 10 – Top view of the two-tile LinkConnect assembly. The waffle-pattern grid is visible across the plate surface, with Clip components at user-selected positions. The partial cover plate (top right) shows the perforated top surface. Rails lock the clips in place along the long tile edges.

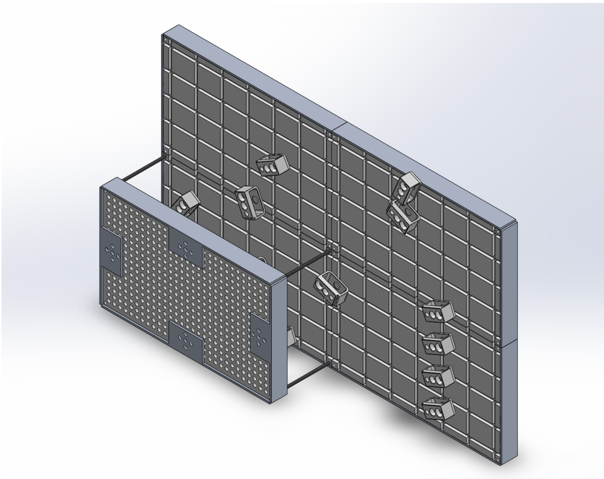

Figure 11 – LinkConnect in a vertical wall-mount configuration, showing the four-tile assembly and the heavily perforated bottom face of the plate. The perforations reduce mass and promote passive convective airflow past cables and attached devices.

pin presses against the Aluminium rail wall under increasing load. Table 3 summarises representative steps from the sweep.

Table 3 – Selected results from the structural load sweep on the clip-pin interface. ABS yield = 44 MPa, Al 1060 yield = 70 MPa.

λ	Displ. (mm)	Pin reac. (N)	Pin reac. (kgf)	ABS status	$\sigma_{vM,\mathrm{ABS}}$ (MPa)
0.00	0.00	0.0	0.0	Safe	0.0
0.20	0.50	485.8	49.5	Safe	14.8
0.40	1.00	964.2	98.3	Safe	35.8
0.50	1.25	1186.5	121.0	Safe	43.2
0.60	1.50	1398.7	142.6	Caution	43.8
0.70	1.75	1598.3	163.0	Yield	44.0
0.80	2.00	1784.2	181.9	Yielding	72.5
0.90	2.25	1956.8	199.5	Yielding	85.3
1.00	2.50	2116.4	215.8	Yielding	96.5

The ABS pin begins to yield at $\lambda \approx 0.70$, while the Aluminium wall remains below its 70 MPa yield threshold across the entire sweep. The pin is therefore the first component to fail under overload, which is by design. Because all components are available individually, a user whose pin yields under extreme load simply replaces that single part, either by printing a replacement at home or ordering one. The rest of the assembly is unaffected. This directly implements the right-to-repair principle central to the LinkConnect design.

6.2 Material Selection

Material selection was driven by the dual requirements of home printability and structural adequacy. Two candidates were evaluated: ABS plastic and Aluminium 6061. Table 4 presents the comparison.

Table 4 – Material comparison for LinkConnect components.

Criterion	ABS Plastic	Aluminium 6061
Yield strength	44 MPa	70 MPa
Young's modulus	2.35 GPa	69 GPa
Density	$1060\,\mathrm{kg\,m^{-3}}$	$2705\,\mathrm{kg\,m^{-3}}$
Heat resistance	$< 90\,°\mathrm{C}$	$> 200\,°\mathrm{C}$
Home 3D printable	Yes (FDM)	No
Recommended infill	100%	N/A (cast)
Fume safety at home	Caution (styrene fumes)	Safe
Recyclability	Code 9 plastic	Infinitely recyclable
Manufacturing process	Injection moulding / FDM	Die casting / extrusion
Component assignment	Plate, Clip, backclip, Pin	Rail_A, Rail_B

For home printing, **PLA** is recommended over ABS. PLA produces negligible fumes and is derived from renewable plant-based feedstocks, making it safe for enclosed home environments. ABS produces styrene vapours during printing, classified as Group 2B by the International Agency for Research on Cancer (IARC), and should only be used with adequate ventilation or an enclosed filtered printer. ABS remains the material of choice for the manufactured production variant due to its superior mechanical performance and injection moulding compatibility.

The FEA assumes **100% infill** throughout all printed parts. Users printing at lower infill should reduce the number of cables per module accordingly to maintain safe structural margins.

6.3 Finite Element Analysis

The structural simulation models the clip-pin contact region as a two-domain plane-strain problem. The ABS pin presses horizontally against the Aluminium rail wall under a prescribed displacement increasing from 0 to 2.5 mm across 80 load steps. Contact between the two domains is enforced at their shared interface. The left boundary of the ABS domain is displaced, and the right boundary of the Aluminium wall is fully fixed. Table 5 gives the material properties used.

Table 5 – Material properties used in the structural simulation.

Material	E (GPa)	ν	ρ (kg/m^3)	σ_Y (MPa)
ABS	2.35	0.37	1060	44
Al 1060	69.0	0.33	2705	70

The key result is that the ABS pin yields at $\lambda \approx 0.70$ while the Aluminium wall never yields. Stress concentrations are localised to the contact interface and do not propagate through the bulk of the pin below $\lambda = 0.85$. In a fully Aluminium assembly, neither component would yield within the simulated range, confirming that the Aluminium variant provides a substantially larger structural safety margin for heavy-duty deployments.

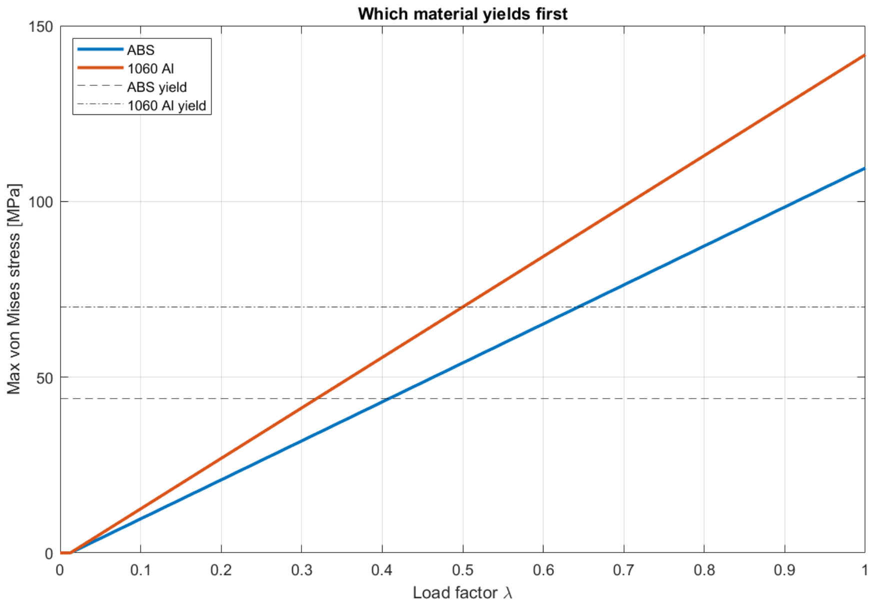

Figure 12 – Maximum von Mises stress versus load factor λ for ABS and Al 1060. The ABS yield threshold (44 MPa, dashed) is crossed at $\lambda \approx 0.70$. The Aluminium threshold (70 MPa, dash-dot) is never reached, confirming the pin as the sacrificial component.

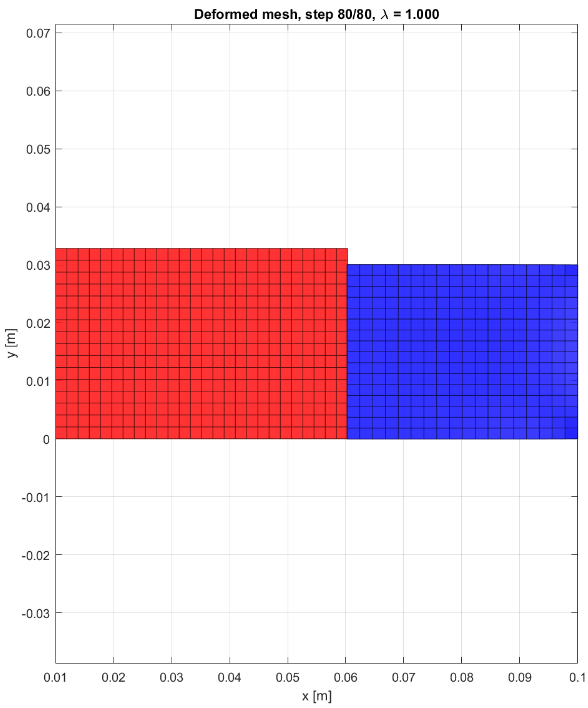

Figure 13 – Deformed mesh at full load (λ = 1.0). ABS domain is shown in red and Aluminium in blue, with colour intensity proportional to local von Mises stress. Stress is concentrated at the contact interface, consistent with expectation.

6.4 Airflow and Thermal Analysis

6.4.1 Plate Geometry and Surface Area

The plate measures 18 cm × 30 cm × 28 mm thick. The bottom face carries 493 perforations of 6 mm diameter and 15 mm depth. These perforations add 0.1395 m^2 of hole surface area to the 0.108 m^2 of flat face area, giving a total effective cooling surface of 0.2475 m^2. This represents a **129% increase** in cooling surface area while removing only **13.8%** of the plate volume: a highly efficient geometric trade-off.

6.4.2 Thermal Simulation

A lumped-capacitance model was used to solve for the steady-state bulk temperature of the plate under a uniform 100 W heat load, accounting for both natural convection and surface radiation at 25 °C ambient. Both ABS and Aluminium were evaluated in perforated and unperforated configurations. Results are summarised in Table 6.

Table 6 – Steady-state temperature and thermal time constant for ABS and Al 6061 under 100 W uniform heat load, 25 °C ambient.

Material	**Configuration**	T_{ss} (°C)	ΔT (°C)	τ **(min)**	**Verdict**
ABS	Solid (no holes)	95.5	+36.0	–	Exceeds HDT – unsafe
ABS	Perforated	59.5	baseline	23.7	Safe
Al 6061	Perforated	72.3	+12.8	30.3	Safe
Al 6061	Solid (no holes)	130.0	+70.5	–	Unsafe – avoid

Several important findings emerge. The unperforated ABS plate reaches 95.5 °C, which exceeds the ABS heat distortion temperature of approximately 90 °C. The perforated version drops to 59.5 °C, a reduction of 36 °C, making the perforations thermally mandatory rather than merely decorative. Despite aluminium having a thermal conductivity of 167 $W\,m^{-1}\,K^{-1}$ versus 0.18 $W\,m^{-1}\,K^{-1}$ for ABS, the perforated ABS plate actually runs cooler in steady state. This is because ABS has an emissivity of $\varepsilon = 0.95$ (near-blackbody), making it a far more effective radiator than Aluminium ($\varepsilon = 0.20$). At the temperatures involved, radiation dominates convection, and ABS exploits this significantly better. Aluminium does however heat up more slowly, with a time constant of 30.3 min versus 23.7 min for ABS, meaning it stays cooler for longer under intermittent or short-duration loads.

6.4.3 Material Recommendation by Use Case

For **cable management**, ABS is the preferred material. It is an electrical insulator, eliminating any risk of short circuits from bare conductors contacting the plate. Its low thermal conductivity acts as a thermal barrier between the plate and the cable jackets, and its high emissivity keeps the surface temperature low through radiation. The perforated ABS plate runs at 59.5 °C under maximum load, which is safe for both the material and the cables it supports.

For **device mounting** applications where power bricks, compact switches, or similar equipment are fixed directly to the plate, the Aluminium variant is preferred. Its high conductivity spreads device-generated heat uniformly across the plate, preventing localised hotspots. Devices rated for elevated base temperatures benefit from this spreading effect, with the perforations then dissipating the distributed heat into the surrounding air.

The production configuration – ABS plate and clips with Aluminium rails – is thermally well balanced. The ABS body provides electrical insulation and effective radiative cooling, while the Aluminium rails conduct heat away from the tile edges, reducing peak temperatures in the centre of the plate.

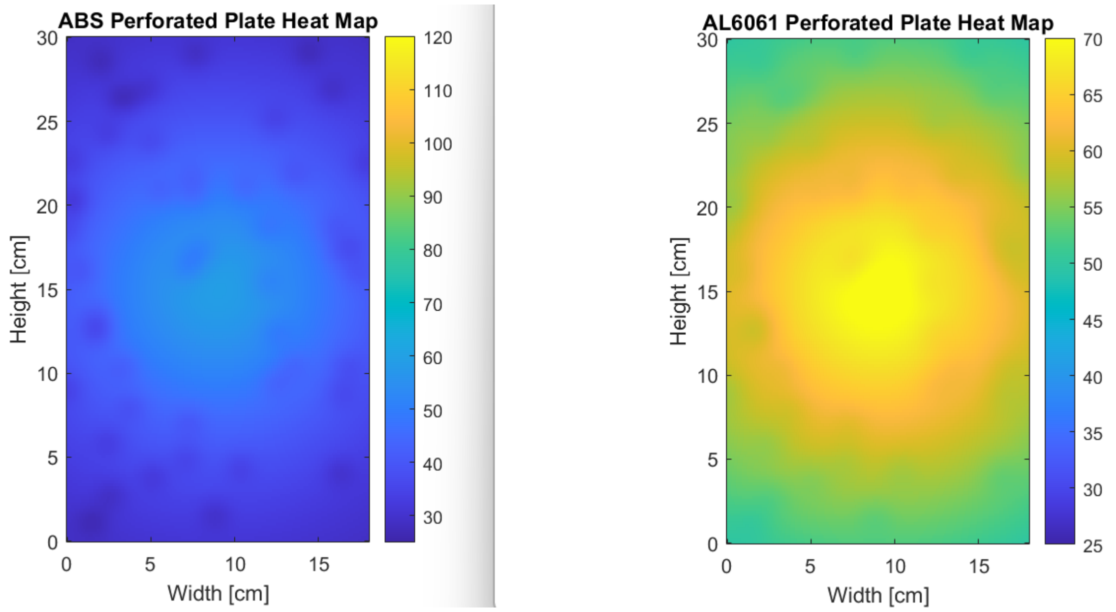

Figure 14 – Simulated spatial temperature distribution for ABS (left) and Al 6061 (right) perforated plates under 100 W load. ABS shows a sharper central hotspot due to poor in-plane heat spreading; Aluminium shows a flatter, more uniform distribution. Individual perforations create localised cooling spots visible across both maps.

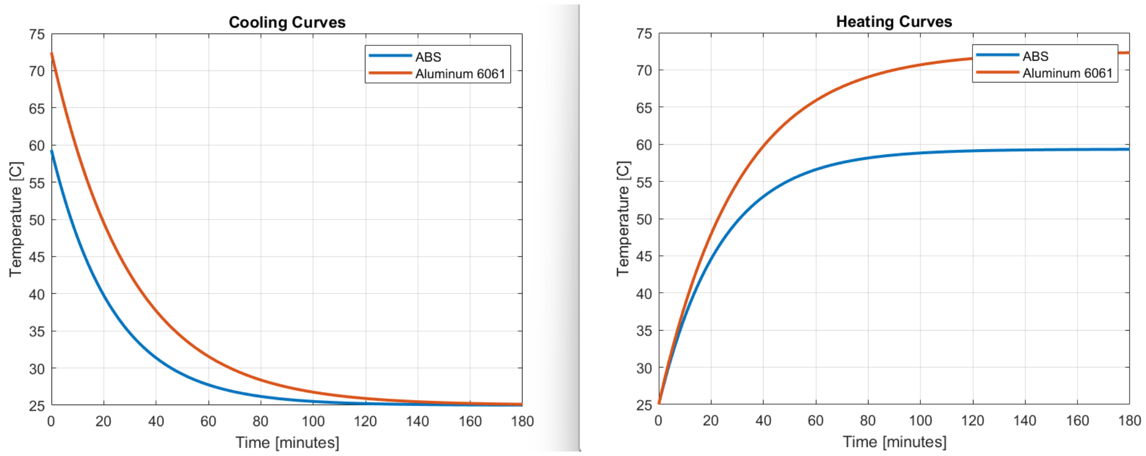

Figure 15 – Transient heating and cooling curves for ABS and Al 6061. ABS reaches steady state faster ($\tau = 23.7$ min) at a lower temperature (59.5 °C). Aluminium heats more slowly ($\tau = 30.3$ min) but equilibrates higher (72.3 °C) due to its low emissivity.

7 Distribution & Community Engagement

7.1 Open-Source Release

When the campus moved to an online environment, the project pivoted from centralised prototyping to digital design and distributed fabrication. Rather than manufacturing and selling physical units, the LinkConnect CAD package is released through open-source repositories such as **GrabCAD** and similar maker platforms, where remote professionals, hobbyists, and engineers can download the SolidWorks files and print components on personal or local 3D printers. The release package contains the complete top-level assembly and all seven individual parts, together with documentation describing the design intent, recommended print parameters, and assembly method. Figure 16 shows the GrabCAD listing for the project.

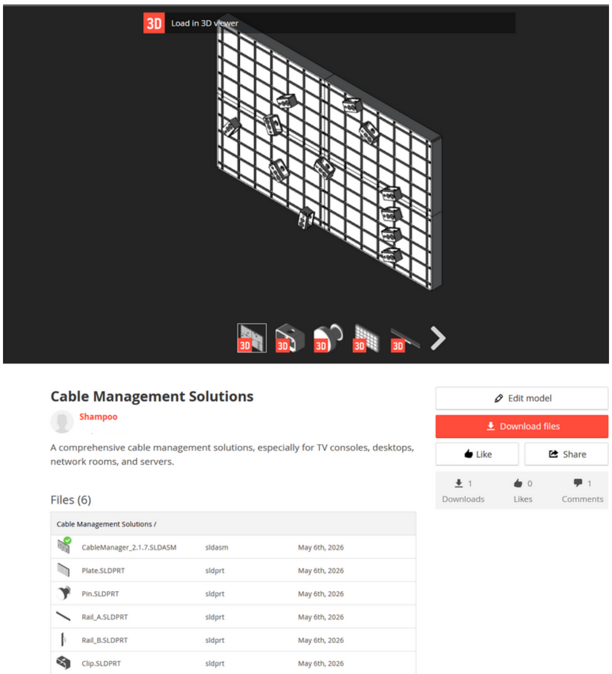

Figure 16 – Open-source release of LinkConnect on GrabCAD, showing the embedded 3D viewer, direct download link, and the complete file package available to the public.

7.2 Iterative Feedback Loop

The open-source release model creates a continuous feedback loop. Users who download and print the modules can leave comments, attach files such as modified parts or photos of printed assemblies, and suggest improvements that can be folded back into the next design revision. An early community comment already proposed adding features for stacking multiple levels of tiles, which is consistent with the modular philosophy of the project and is being considered for the next revision. Rather than being frozen at release, the design evolves continuously as users adapt it to different cable diameters, desk geometries, and mounting methods. Figure 17 shows the project file listing and the first community comment received.

7.3 Potential Uses & Buyers

Beyond the home and small-office market, two additional categories of users were identified. The first is **office furniture manufacturers**, who can integrate LinkConnect tiles directly into the underside of new desks: the parametric tile geometry can be adjusted to match any desk footprint and shipped as a pre-installed accessory. The second is **server and networking equipment vendors**, such as large industrial automation companies and network infrastructure providers. In a server room or network rack, the discrete-channel approach organises high-density cable bundles while preserving the airflow critical to thermal management. Figure 18 shows a SolidWorks integration of LinkConnect into the underside of a desk, alongside example end-customer contexts.

8 Market Analysis & Commercial Viability

8.1 Overview

A common concern with niche hardware products is that the potential customer base may be too narrow to sustain a viable commercial operation. This section addresses that concern directly. **Cable management**

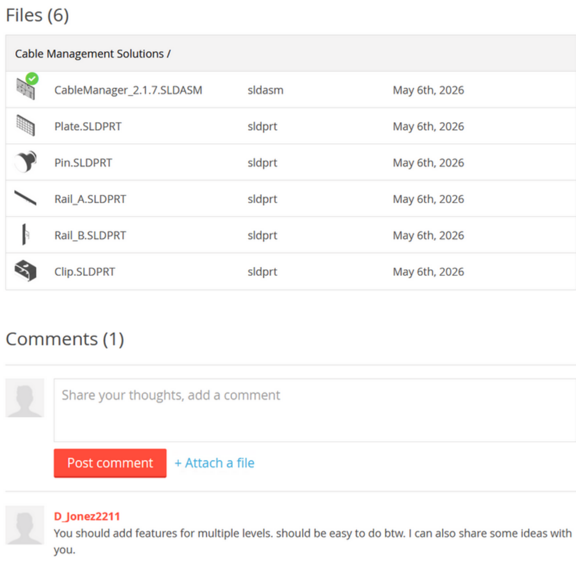

Figure 17 – Iterative feedback loop on GrabCAD. The file listing (left) shows all downloadable parts, and the first community comment proposes multi-level tile stacking. The rendered assemblies (right) show both horizontal under-desk and vertical wall-mount configurations.

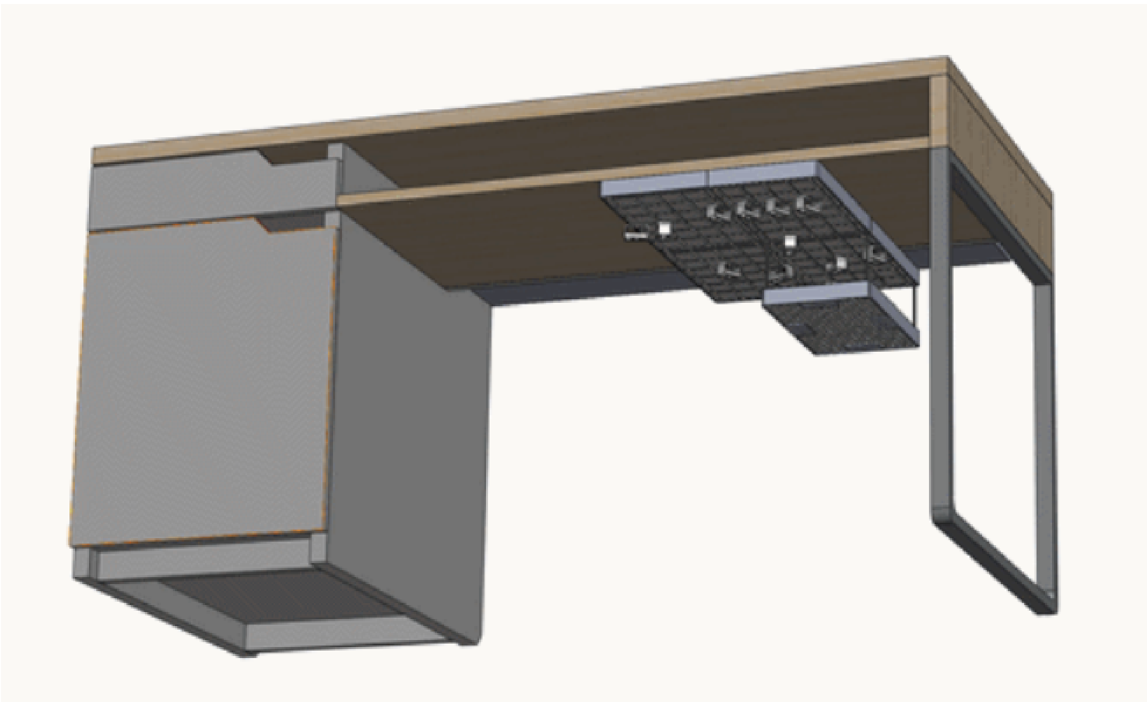

Figure 18 – Potential uses and buyers. A desk with LinkConnect integrated under the work surface (right), alongside example industrial and network equipment vendor contexts (left).

is not a niche problem. Every individual, household, office, school, hospital, data centre, and industrial facility that uses electronic equipment faces the same problem LinkConnect solves. The market is not limited; it is universal.

8.2 Global Cable Management Market

The global cable management market was valued at approximately **USD 18.7 billion** in 2023 and is projected to grow at a CAGR of approximately 12% through 2030, driven by the proliferation of electronic devices, smart home adoption, data centre expansion, and the global shift toward hybrid and remote work. The market spans four key segments: residential (home offices, smart home installations), commercial (corporate offices, co-working spaces), industrial (server rooms, data centres, automation systems), and institutional (schools, universities, hospitals). Existing products – cable trays, Velcro sleeves, cable boxes, and zip ties – address these segments, but none offer the combination of *modularity, structural separation, repairability, and open-source accessibility* that LinkConnect provides.

8.3 UAE Market Context

The UAE presents a particularly strong launch environment for LinkConnect. The **UAE Vision 2031** and **UAE Digital Economy Strategy** target doubling the digital economy's contribution to GDP from 9.7% to over 19.4% within a decade, driving massive investment in data centres and smart offices. The UAE is already home to over 20 operational data centres, with further expansion planned in Abu Dhabi and Dubai by hyperscalers including Microsoft Azure, AWS, and Oracle Cloud. On the residential side, over 43,000 new units were delivered in Dubai in 2023 alone, with smart home technology increasingly standard in new developments. The UAE consumer electronics market was valued at approximately **USD 3.4 billion** in 2023, with Dubai's retail infrastructure (Sharaf DG, Jumbo Electronics, Amazon.ae, Noon.com) providing ready distribution channels.

The UAE innovation ecosystem further supports a hardware startup like LinkConnect. Institutions including the **Dubai Future Foundation**, **Hub71** (Abu Dhabi), **Dubai Silicon Oasis**, **twofour54**, and the **Sharjah Research Technology and Innovation Park (SRTIP)** provide funding, maker spaces, 3D printing infrastructure, and accelerator programmes through which the product could scale rapidly.

8.4 Competitive Analysis

Table 7 presents a structured comparison of LinkConnect against the four most common existing cable management solutions.

Table 7 – Competitive analysis of LinkConnect versus existing cable management solutions.

Criterion	LinkConnect	Cable Tray	Velcro Sleeve	Cable Box	Zip Ties
Separates individual cables	✓	×	×	×	×
Modular / expandable	✓	Limited	×	×	×
Repairable at home	✓	×	Limited	×	×
Open-source / free to print	✓	×	×	×	×
Airflow preservation	✓	Limited	×	×	Limited
Structurally validated (FEA)	✓	×	×	×	×
Scalable to full workstation	✓	Limited	×	×	×
Easy cable add/remove	✓	Limited	Limited	×	×
Industrial grade variant	✓	✓	×	×	×

No existing product satisfies more than three of the nine criteria simultaneously. LinkConnect satisfies all nine. The four key differentiators are true cable separation (each cable in its own discrete channel, not bundled), unlimited scalability (a single tile can expand to cover an entire desk, wall, or rack – **every cable from every device routed through one unified system**), right to repair (individual components replaced by printing at home, aligned with the EU Right to Repair Directive 2024), and open-source accessibility (free to download and print, generating organic adoption without marketing spend).

8.5 Target Customer Segments

8.5.1 Home Office and Remote Workers

The global remote workforce exceeded 1.5 billion people in 2023, with approximately 30% of UAE private sector workers in hybrid arrangements post-pandemic. A typical home office generates 6–15 cables from 4–8 devices. LinkConnect organises all of them for under AED 37 in printed materials, cheaper than most commercial solutions at Sharaf DG, with significantly more functionality. The primary marketing channels for this segment are social media (Instagram, TikTok, YouTube) and maker communities on GrabCAD and Reddit.

8.5.2 PC Enthusiasts and Gamers

The global gaming peripherals market exceeded USD 5 billion in 2023, with over 3 million active gamers in the UAE alone. Gaming setups are cable-dense and cable management is a visible point of pride in this community. LinkConnect's modular aesthetic and ability to route every peripheral through a single unified surface is a direct fit for the battlestation culture. YouTube tech reviewers, Instagram setup tours, and Maker Faire demonstrations are the key channels here.

8.5.3 Small and Medium Enterprises

The UAE has over 557,000 registered SMEs contributing 63% of non-oil GDP. Most SME offices contain 5–50 workstations generating significant cable clutter. Commercial cable trays are expensive to fit and inflexible when layouts change. LinkConnect tiles deploy without tools, reposition when desks move, and at AED 17 per manufactured unit make full-office deployment affordable. B2B sales via LinkedIn and partnerships with UAE co-working operators (Astrolabs, In5, WeWork) are the primary routes to this segment.

8.5.4 Educational Institutions and Data Centres

The UAE has over 1,100 schools, 70+ universities, and a rapidly growing data centre market valued at USD 3.1 billion in 2023 (CAGR 14%). Computer labs and server rooms share the same problem: extreme cable density where individual cable access is frequently needed. The discrete-channel design makes cable identification and replacement straightforward, while the Aluminium variant provides the structural and thermal performance required for rack-mounted deployments. Potential enterprise clients in the UAE include Khazna Data Centres, du, and Etisalat (e&). The primary channels are institutional procurement, Ministry of Education partnerships, and GITEX Technology Week.

8.5.5 Furniture Manufacturers and OEM Integration

Premium office furniture manufacturers (IKEA, Herman Miller, Steelcase) increasingly compete on integrated cable management. LinkConnect's parametric design can be adapted to any desk footprint and shipped as a factory-installed accessory under an OEM licence, requiring no retail infrastructure on LinkConnect's part. Design Days Dubai and the Index Exhibition are the primary channels for this segment.

8.6 Go-to-Market Strategy

The recommended strategy follows three phases. In **Phase 1 (Months 0–6)**, the GrabCAD release and maker communities drive organic awareness at near-zero cost through Reddit, YouTube, and Instagram engagement. In **Phase 2 (Months 6–18)**, a manufactured product launches through UAE retail channels with support from Hub71 or the Dubai Future Accelerators, with an estimated investment of AED 150,000–250,000. GITEX Technology Week provides the anchor exhibition opportunity. In **Phase 3 (Months 18–36)**, OEM licensing is pursued with furniture manufacturers and data centre operators, manufacturing is scaled through injection moulding partners in Jebel Ali Free Zone (JAFZA), and the product expands into GCC markets.

8.7 Addressing the Concern of Limited Customers

The evidence above directly refutes the concern that potential customers may be limited. The global cable management market exceeds USD 18 billion annually. The UAE alone contains over 557,000 SMEs, 1,100+ schools, 20+ data centres, and 43,000+ new residential units per year. The open-source model removes the price barrier entirely, and the OEM furniture route provides a path to volume without retail infrastructure. LinkConnect is not a product in search of customers. It solves a problem that exists in every home, office, school, and data centre in the world. The challenge is not finding customers; it is deciding which segment to serve first.

9 Conclusion

9.1 Summary of Accomplishments

LinkConnect set out to address cable clutter at modern workstations with a structurally rigorous, parametric, and openly distributed solution. **Every deliverable proposed at the start of the project was completed**:

- A fully parametric SolidWorks assembly comprising seven interacting parts (Plate, Clip, backclip, Rail_A, Rail_B, SeparatingRod, Pin).
- Quantitative structural analysis via load estimation and an 80-step FEA load sweep, identifying the pin as the sacrificial component and confirming the Aluminium variant as safe across the full load range.
- A separate thermal simulation validating the perforated plate design and providing material recommendations by use case.
- Material selection and validation for both ABS and Aluminium, covering home fabrication, mass manufacturing, and environmental impact.
- A public open-source release on GrabCAD with full documentation, an embedded 3D viewer, and an active community feedback channel that received its first contribution within one week.

The combination of discrete cable pathing, modular interlocking tiles, and distributed fabrication yields a system that is more mechanically informed than existing cable trays and Velcro sleeves, while remaining accessible to anyone with a consumer-level 3D printer.

9.2 Design Discussion

9.2.1 Design Validation and Implications

The parametric CAD approach proved highly effective for iterative refinement. By tying all geometric dimensions to sketch parameters, the design could rapidly explore trade-offs between cable spacing, wall thickness, and material efficiency without rebuilding geometry from scratch. The discrete three-hole clip geometry emerged as the most robust solution: it provides sufficient separation between adjacent cables to prevent tangling, while fitting within the compact footprint required for under-desk mounting. The interlocking rail system proved mechanically sound in the FEA, with stress concentrations confined to the expected regions at the clip-pin contact interface. This gives confidence that the design can scale to larger tiles or alternative mounting orientations without fundamental rethinking.

9.2.2 Structural Findings and Safety Margins

The load sweep revealed an important material boundary: ABS yields at approximately $\lambda = 0.70$, meaning the ABS variant should be used with a reduced cable count of roughly 60–70% of the nominal eight to fifteen cables per module. This is not a design flaw but a natural constraint of additive manufacturing. 3D-printed parts are lighter and cheaper than machined equivalents, but their structural limits must be respected. The pin is intentionally the weakest component: it yields first and can be individually replaced at home without discarding the rest of the assembly. The Aluminium variant provides a large safety margin across the full load range, making it the preferred choice for high-traffic or industrial environments. Future iterations could explore hybrid constructions combining ABS clips with Aluminium rails to balance cost and performance.

9.2.3 Thermal Discussion

The thermal simulation confirmed that the 493 perforations on the bottom face of the plate are not merely aesthetic; they are structurally necessary for the ABS variant to remain within its safe operating temperature. Without perforations, the ABS plate reaches 95.5 °C under full load, exceeding its heat distortion temperature. With perforations, it drops to 59.5 °C. Despite aluminium's far superior thermal conductivity, the perforated ABS plate runs cooler in steady state due to its significantly higher emissivity ($\varepsilon = 0.95$ versus 0.20 for Aluminium). Radiation dominates at the temperatures of interest, and ABS exploits this more effectively. The conclusion is that ABS is the correct material for cable management applications, acting as both a thermal barrier and an effective radiator, while Aluminium is recommended where devices are mounted directly to the plate and heat spreading is the priority.

9.2.4 Material Selection Trade-offs

Supporting both ABS and Aluminium reflects the project's philosophy of accessibility. ABS allows anyone with a consumer 3D printer to fabricate the system at home for under AED 37. Aluminium enables larger organisations to deploy a higher-performance variant without redesigning the system. The environmental trade-off is nuanced: ABS printed from recycled feedstock has lower embodied carbon than virgin aluminium, but recycled aluminium requires only 5% of the energy of primary production and is infinitely reusable. The right choice depends on the user's context, which is precisely why both routes are provided and documented.

9.2.5 Comparison with Existing Solutions

LinkConnect addresses the core limitations of the two most common alternatives. Under-desk cable trays reduce visual clutter but collect all cables into a single channel, where they compress against each other, restrict heat dissipation, and amplify mechanical stress at connectors. Velcro sleeves bundle cables even more tightly, making individual cable access difficult and airflow worse. LinkConnect's discrete pathing gives each cable its own channel, preserves airflow through the waffle grid and perforated bottom face, and allows individual cables to be added or removed without disturbing neighbours. The cost premium over generic cable trays is justified by improved thermal performance, easier maintenance, structural validation, and the ability to scale to cover an entire workstation.

9.2.6 Community Engagement and Iteration

The open-source GrabCAD release was motivated by the shift to online learning, which made centralised prototyping infeasible. Rather than treating this as a limitation, the project embraced it as an opportunity. The first community comment, proposing multi-level tile stacking, demonstrates both demand and a clear direction for the next iteration. This feedback loop is fundamentally different from traditional product development, where a static design is manufactured and sold. Here the design evolves continuously as users adapt it, and every improvement benefits the entire community at no additional cost.

9.3 Lessons Learned

1. **Parametric design is essential.** Hard-coded dimensions produce brittle designs that break when requirements change. Parametric features allowed rapid exploration of cable diameters, grid spacing, and material choices without rebuilding any part from scratch.
2. **FEA informs but does not replace engineering judgment.** The load sweep provided quantitative validation, but deciding what the results mean – choosing a safety factor, determining when ABS is acceptable, identifying the pin as the sacrificial component – required reasoning informed by the end-use context.
3. **Simulation and theory should complement each other.** Both the structural FEA and the thermal lumped-capacitance model produced results that were non-obvious: ABS runs cooler than Aluminium in steady state due to emissivity, and ABS yields at the pin long before the Aluminium wall is affected. Neither result would have been confidently predicted without the simulation.
4. **Open-source distribution has real, immediate impact.** Uploading the design to GrabCAD required minimal effort yet generated community engagement within one week. Removing barriers to access is a force multiplier for any engineering project.
5. **Material selection is a system-level decision.** Choosing between ABS and Aluminium was not binary. The project identified a two-tier approach – ABS for home users, Aluminium for industry – that serves both market segments without compromising either.
6. **Iterate early and often.** The three-hole clip geometry was not the first concept explored. Multiple sketches were built and tested before arriving at a design that balanced cable separation, compact footprint, and manufacturing feasibility.

9.4 Future Work

1. **Multi-level stacking.** The first GrabCAD community comment proposed stacking tiles vertically for server racks, wall-mounted displays, and large workstations. The modular design already supports this with minimal rail geometry modifications.
2. **Parametric cable-diameter presets.** Pre-built clip configurations for common cable types (IEC C7, Cat5e, Cat6, Cat6a, USB-C, Thunderbolt) so users select a preset and all geometry updates automatically.
3. **Hybrid material constructions.** ABS clips with Aluminium rails offer a cost-performance middle ground suited to commercial furniture makers. This is the natural next step between the home and industrial variants.
4. **Aluminium server-room variant.** A fully Aluminium design optimised for 19-inch rack mounting and hot-swap cable management in data centres, targeting infrastructure vendors in the UAE and GCC region.
5. **CFD thermal validation.** A computational fluid dynamics study to quantify the airflow improvement of the waffle grid and perforated bottom over bundled-cable and solid-tray alternatives, providing data to support commercial claims.
6. **Community expansion.** Broader engagement across maker platforms to grow the contributor base, collect printed assembly photos, and drive continuous design improvement.

9.5 Closing Reflection

This project demonstrated that engineering rigour and open-source philosophy are complementary. A parametric design validated through structural and thermal simulation, then released under an open licence, does not compromise quality; it broadens accessibility and enables a global feedback loop that centralised product development cannot match. The modular tile concept proved to be a powerful organising principle, simplifying manufacturing, scaling, and community contribution simultaneously. The two-tier material strategy shows that a single design can serve multiple market segments without becoming a lowest-common-denominator compromise.

Addressing a mundane problem – cable clutter – with modern engineering tools (parametric CAD, FEA, thermal simulation, open-source distribution) yields a solution that is both practically useful and technically rigorous. LinkConnect is not a breakthrough in materials science or a novel structural principle. It is a well-executed application of existing engineering knowledge to a real, everyday problem, multiplied by the power of open-source to let that solution scale globally.

10 References

Gibson, I., Rosen, B., and Stucker, B. (2014). *Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing*. Springer, 2nd edition.

Zienkiewicz, O. C. (1977). *The Finite Element Method in Engineering Science*. McGraw-Hill.

Bonwetsch, F., Shea, K., and Stouffs, R. (2006). Parametric structural design and its relationship to nonlinear finite element analysis. *International Journal of Architectural Computing*, 4:27–42.

Bak, D. (2010). Rapid manufacturing for wound care: A review of the state-of-the-art. *Advanced Drug Delivery Reviews*, 62:1172–1182.

Stallman, R. M. (2014). *Free Software, Free Society: Selected Essays of Richard M. Stallman*. Free Software Foundation.

Kelty, C. M. (2008). *Two Bits: The Cultural Significance of Free Software*. Duke University Press.

Boothroyd, G., Dewhurst, P., and Knight, W. A. (2002). *Product Design for Manufacture and Assembly*. Marcel Dekker Inc., 2nd edition.

Wohlers, T. (2021). *Wohlers Report 2021: 3D Printing and Additive Manufacturing Global Assessment*. Wohlers Associates.

Ellen MacArthur Foundation (2019). Circular economy and electronics: Design for the circular economy. Ellen MacArthur Foundation.

European Commission (2024). Right to repair directive 2024: Consumer rights and sustainability. *Official Journal of the European Union*.

Incropera, F. P., DeWitt, D. P., Bergman, T. L., and Lavine, A. S. (2020). *Fundamentals of Heat and Mass Transfer*. John Wiley & Sons, 8th edition.

Oshwa (2019). Open source hardware association definition and certification. Open Source Hardware Association.

A Engineering Drawings

This appendix contains the eight SolidWorks engineering drawings for all LinkConnect components.

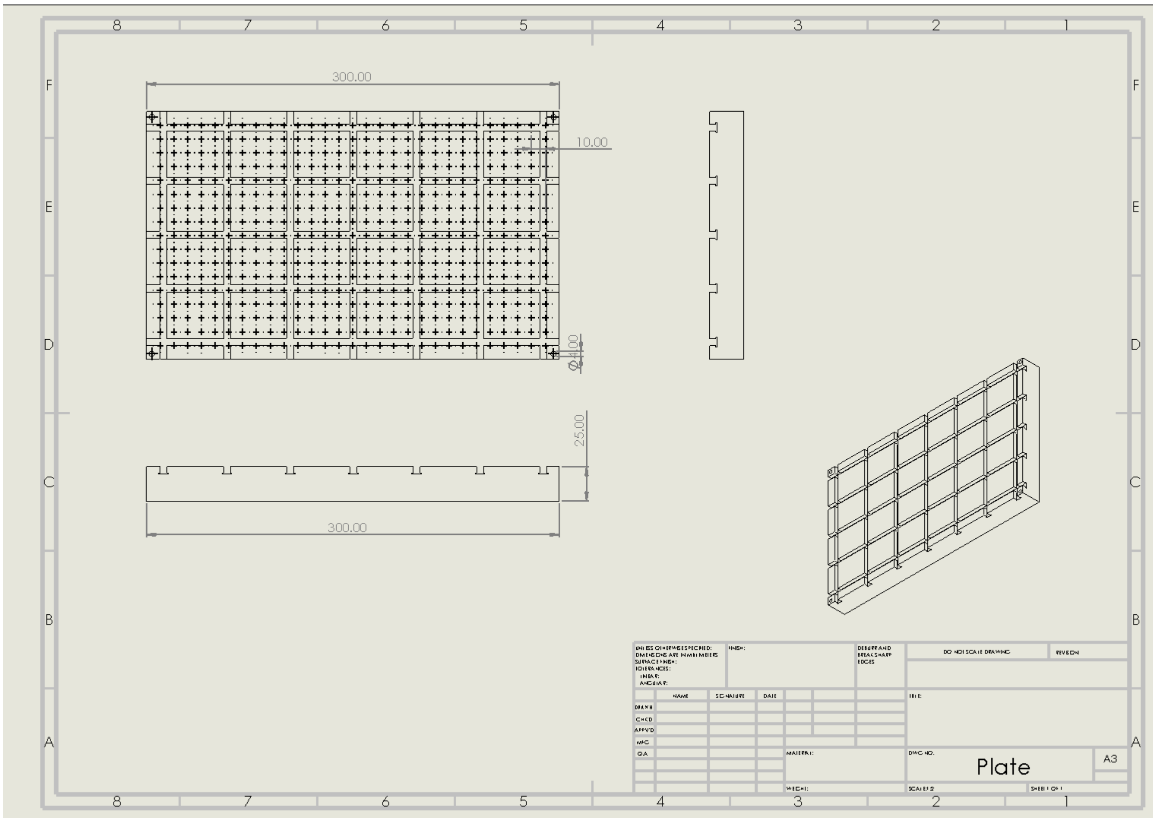

Figure 19 – Engineering drawing of the Plate component.

B MATLAB Code for Structural FEA

The following MATLAB code performs the 80-step finite element load sweep validation of the clip-pin interface for both ABS and Aluminium materials.

```
function WorkingOK()
clearvars; clc; close all;

matABS.name = "ABS"; matABS.E = 2.35e9; matABS.nu = 0.37;
matABS.rho = 1060; matABS.sigY = 44e6; matABS.color = [0.85 0.20 0.20];
matAl.name = "Al1060"; matAl.E = 69e9; matAl.nu = 0.33;
matAl.rho = 2705; matAl.sigY = 70e6; matAl.color = [0.20 0.20 0.85];
matABS.D = DplaneStrain(matABS.E, matABS.nu);
matAl.D = DplaneStrain(matAl.E, matAl.nu);

Lcyl = 0.060; Lwal = 0.040; H = 0.030; nx1 = 26; ny = 16; nx2 = 18;
geom.x0 = 0.0; geom.x1 = Lcyl; geom.x2 = Lcyl + Lwal;
geom.y0 = 0.0; geom.y1 = H;

meshABS = makeStructuredQuadMesh(geom.x0, geom.x1, geom.y0, geom.y1, nx1,
    ny);
meshAl = makeStructuredQuadMesh(geom.x1, geom.x2, geom.y0, geom.y1, nx2, ny
    );
model = mergeTwoDomains(meshABS, meshAl);
```

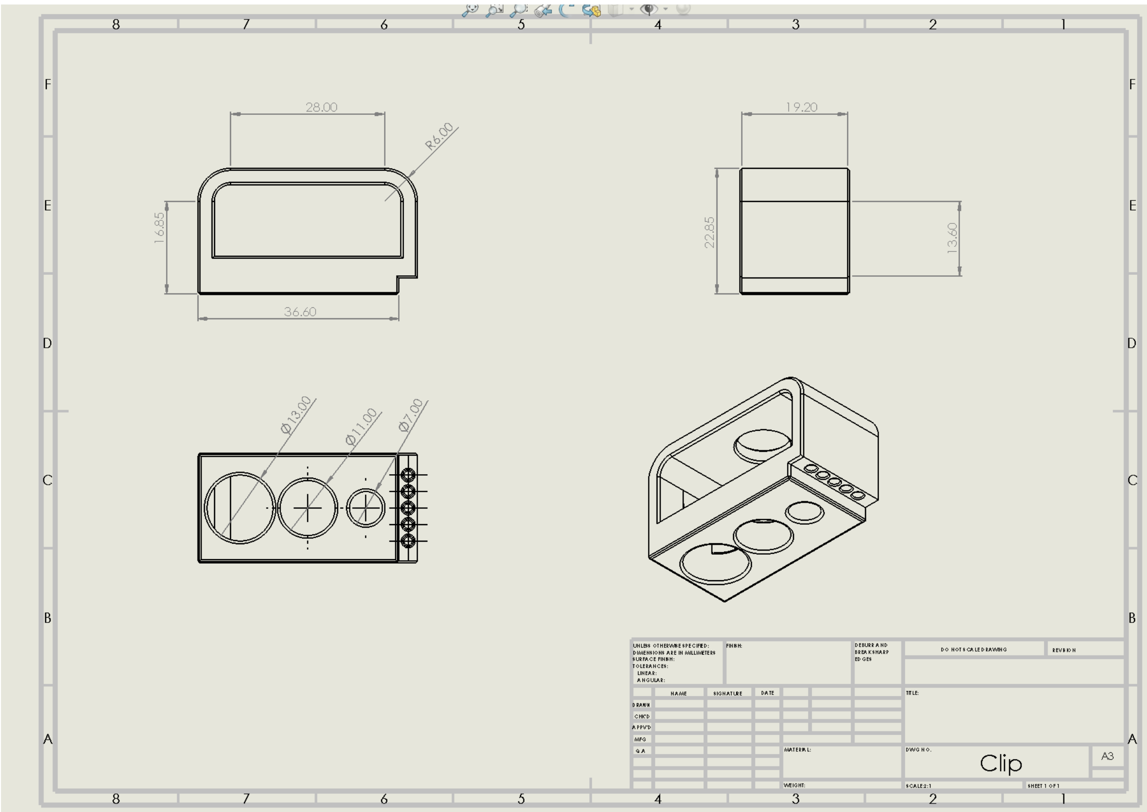

Figure 20 – Engineering drawing of the Clip component.

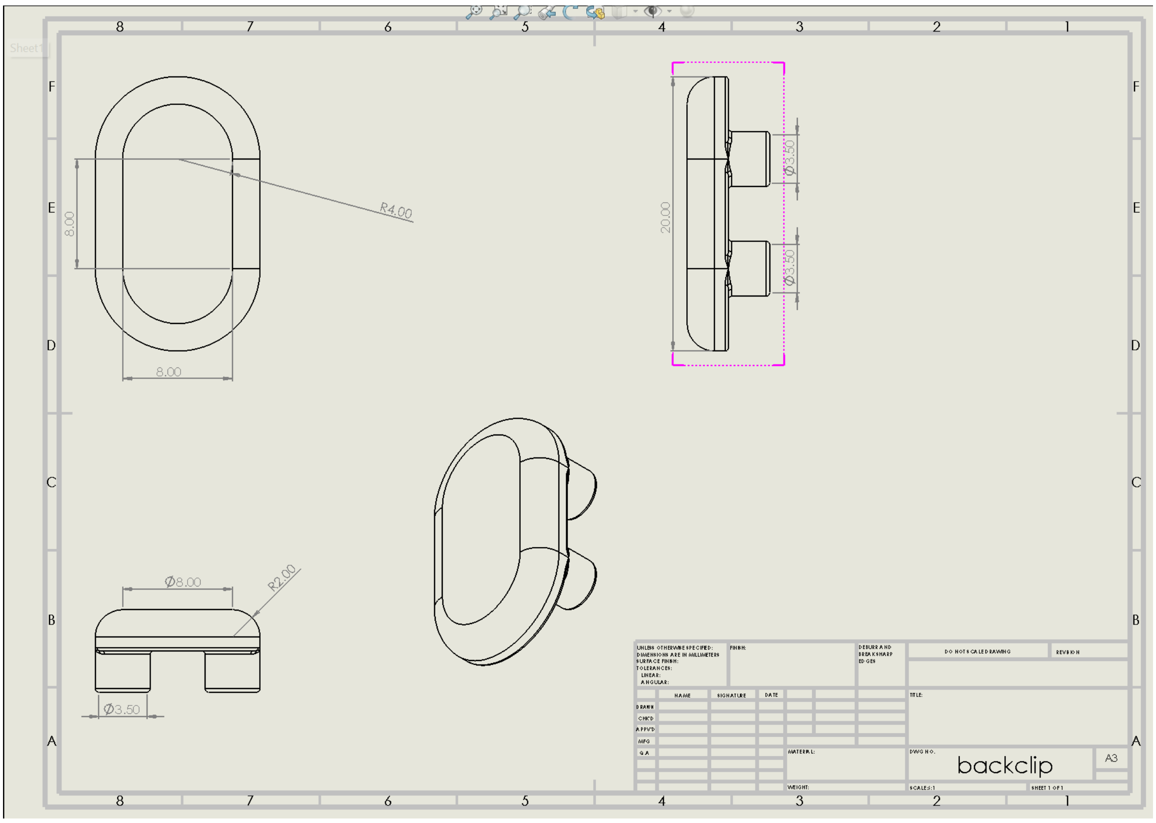

Figure 21 – Engineering drawing of the backclip component.

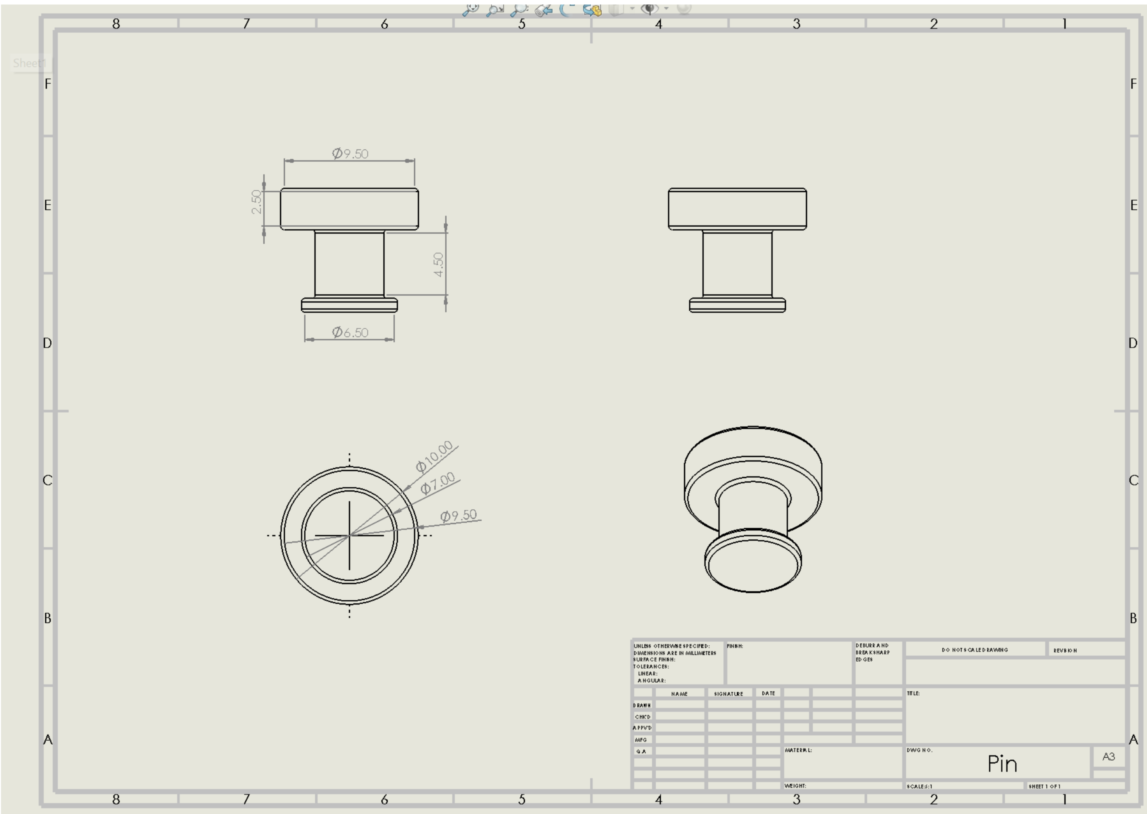

Figure 22 – Engineering drawing of the Pin component.

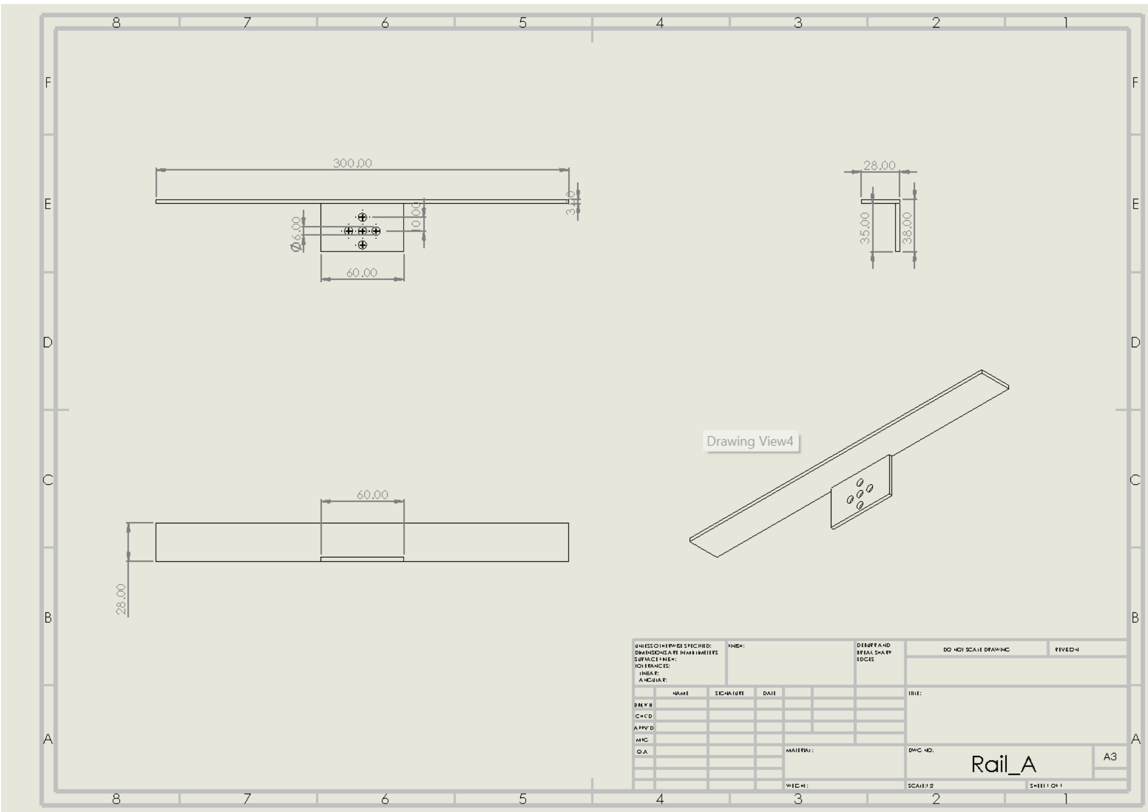

Figure 23 – Engineering drawing of the Rail_A component.

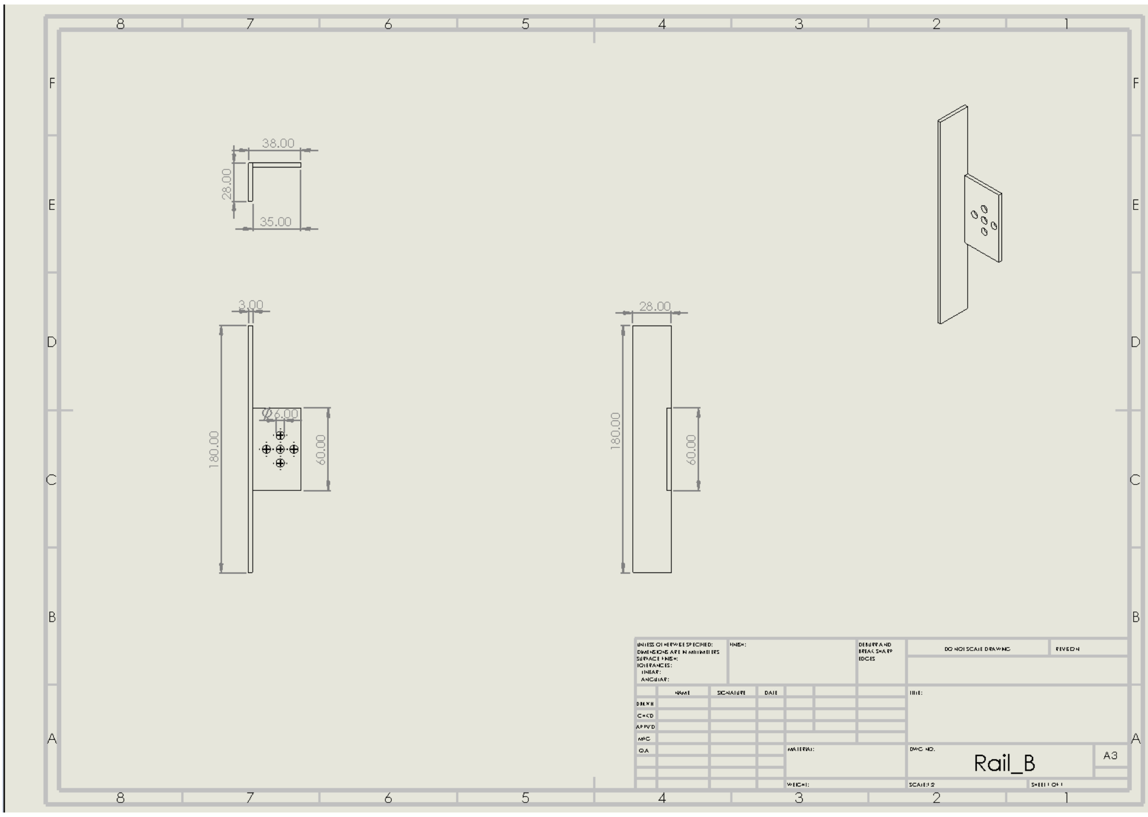

Figure 24 – Engineering drawing of the Rail_B component.

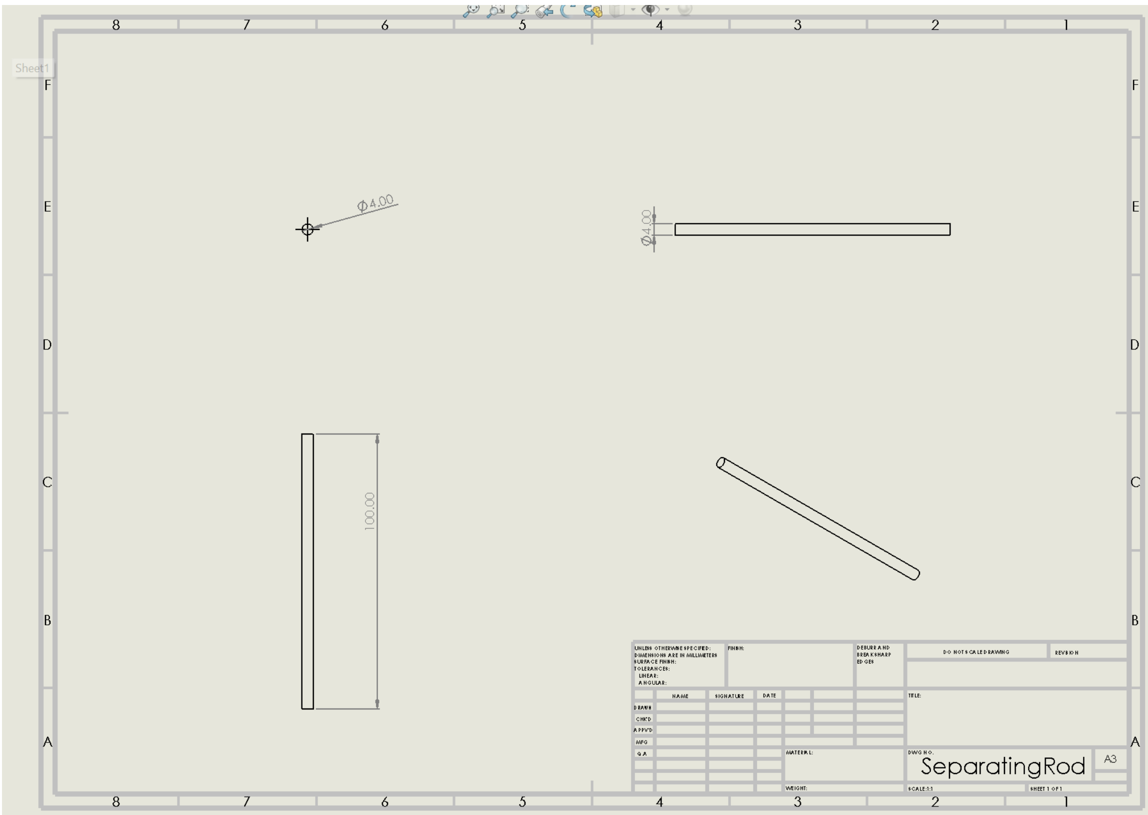

Figure 25 – Engineering drawing of the SeparatingRod component.

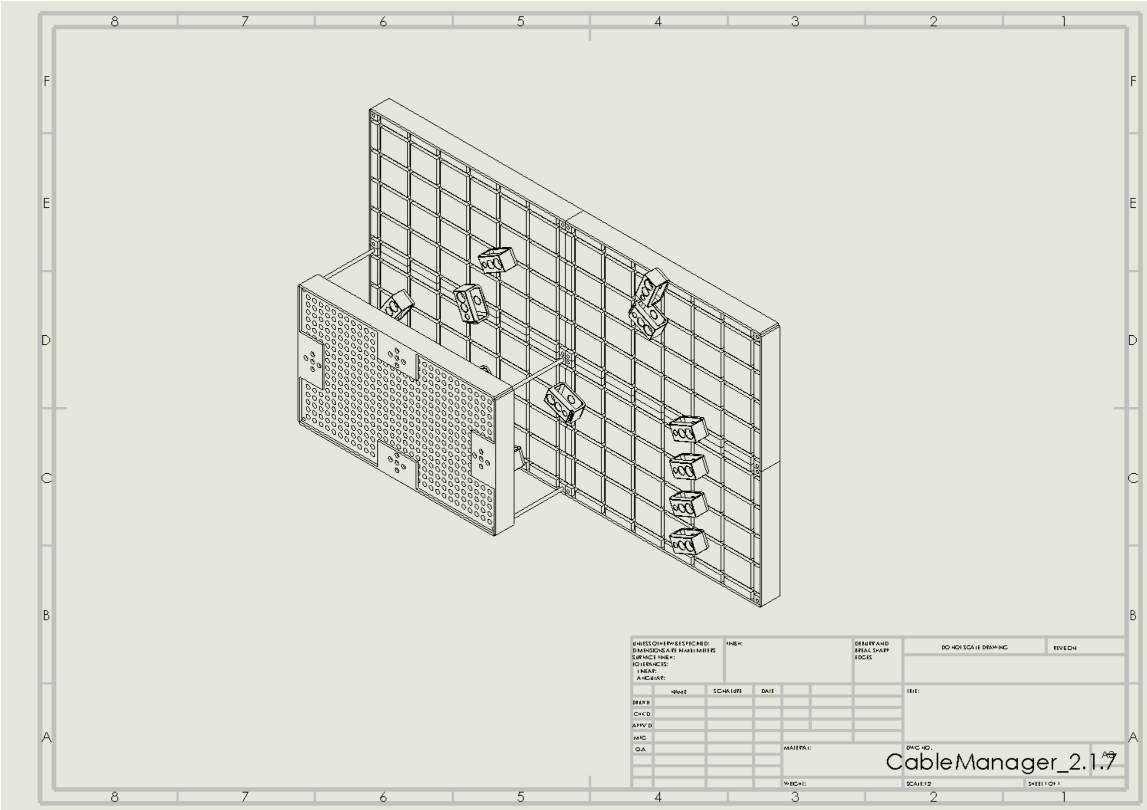

Figure 26 – Engineering drawing of the complete LinkConnect assembly.

```
nA = meshABS.nnode; nB = meshAl.nnode;
bodyID = [ones(nA,1); 2*ones(nB,1)];
pairs = buildInterfacePairs(meshABS, meshAl, model);

fprintf('Assembling stiffness matrix ...\n');
[K, elemData] = assembleGlobalStiffness(model, bodyID, matABS, matAl);
fprintf('Done.\n');
C = 0 * sparse(size(K,1), size(K,2));

leftNodes = model.boundary.leftABS;
rightNodes = model.boundary.rightWall;
uMax = 2.5e-3; nSteps = 80; loadLevels = linspace(0,1,nSteps);
hmin = min(model.hx, model.hy);
kpen = 200 * max(matABS.E, matAl.E) / hmin;

ndof = 2 * model.nnode;
Uhist = zeros(ndof, nSteps);
Rleft = zeros(nSteps,1); Rright = zeros(nSteps,1);
maxVmA = zeros(nSteps,1); maxVmB = zeros(nSteps,1);
Rpin = zeros(nSteps,1);
U = zeros(ndof,1);

vidfile = fullfile(pwd, 'abs_1060_contact_fem_demo.mp4');
vw = VideoWriter(vidfile, 'MPEG-4'); vw.FrameRate = 12; open(vw);
fig = figure('Color','w','Position',[80 80 1350 760]);

fprintf('Running nonlinear contact steps ...\n');
for s = 1:nSteps
```

```
    lam = loadLevels(s);
    uLeft = lam * uMax;
    fixed = false(ndof,1); fixedVal = zeros(ndof,1);
    for i = 1:numel(leftNodes)
        n = leftNodes(i); fixed(2*n-1) = true; fixedVal(2*n-1) = uLeft;
    end
    for i = 1:numel(rightNodes)
        n = rightNodes(i); fixed(2*n-1) = true; fixed(2*n) = true;
    end
    pinNode = model.boundary.pinYNode;
    fixed(2*pinNode) = true;

    U = solveWithPenaltyContact(K, C, model, pairs, kpen, fixed, fixedVal, U);
    Uhist(:,s) = U;
    [R, ~] = computeReactions(K, C, model, pairs, kpen, U, fixed, fixedVal);
    Rleft(s) = sum(R(2*leftNodes-1));
    Rright(s) = sum(R(2*rightNodes-1));
    Rpin(s) = R(2*pinNode);
    post = computeStresses(model, bodyID, elemData, U, matABS, matAl);
    maxVmA(s) = max(post.vmises(model.elemBody == 1));
    maxVmB(s) = max(post.vmises(model.elemBody == 2));

    clf(fig);
    tiledlayout(fig,1,2,'Padding','compact','TileSpacing','compact');
    nexttile(1);
    plotDeformedMeshColored(model, U, post.vmises, matABS, matAl, 4.0);
    title(sprintf('Deformed mesh, step %d/%d, \\lambda = %.3f', s, nSteps, lam));
    xlabel('x [m]'); ylabel('y [m]');
    nexttile(2);
    plot([0 1], [matABS.sigY matABS.sigY]/1e6, 'k--', 'LineWidth', 1.2); hold on;
    plot([0 1], [matAl.sigY matAl.sigY]/1e6, 'k-.', 'LineWidth', 1.2);
    plot(loadLevels(1:s), maxVmA(1:s)/1e6, 'LineWidth', 2);
    plot(loadLevels(1:s), maxVmB(1:s)/1e6, 'LineWidth', 2);
    grid on; box on; xlim([0 1]);
    xlabel('Load factor \lambda'); ylabel('Max von Mises stress [MPa]');
    legend({'ABS yield','Al 1060 yield','ABS max \sigma_{vm}','Al 1060 max \sigma_{vm}'},...
        'Location','northwest');
    title('Yield check');
    drawnow;
    frame = getframe(fig);
    writeVideo(vw, frame);
    fprintf('Step %3d/%3d: maxVmABS = %8.2f MPa, maxVmAl = %8.2f MPa\n',...
        s, nSteps, maxVmA(s)/1e6, maxVmB(s)/1e6);
end
close(vw);
fprintf('Video written to: %s\n', vidfile);

[~, idxA] = find(maxVmA >= matABS.sigY, 1, 'first');
[~, idxB] = find(maxVmB >= matAl.sigY, 1, 'first');
fprintf('\n--- FIRST-YIELD REPORT ---\n');
if isempty(idxA) && isempty(idxB)
    fprintf('Neither material reaches nominal yield.\n');
else
```

```
    if isempty(idxA); idxA = inf; end
    if isempty(idxB); idxB = inf; end
    if idxA < idxB
        fprintf('ABS yields first at step %d, load factor %.4f.\n', idxA, loadLevels(idxA));
    else
        fprintf('Al yields first at step %d, load factor %.4f.\n', idxB, loadLevels(idxB));
    end
end

postFinal = computeStresses(model, bodyID, elemData, Uhist(:,end), matABS, matAl);
vmAll = postFinal.vmises;
elemAbs = find(model.elemBody == 1);
vmAbs = vmAll(elemAbs);

[maxVmAbs, idxMaxAbs] = max(vmAbs);
elemMax = elemAbs(idxMaxAbs);
coordMax = mean(model.nodes(model.elem(elemMax,:), :), 1);

fprintf('\n--- ABS FAILURE DETAILS ---\n');
fprintf('Max von Mises in ABS at final step: %.2f MPa\n', maxVmAbs/1e6);
fprintf('Location: element %d, centroid (%.4f, %.4f) m\n', elemMax, coordMax(1), coordMax(2));

failMask = vmAbs >= matABS.sigY;
failIDs = elemAbs(failMask);
failVm = vmAbs(failMask);
if isempty(failIDs)
    fprintf('No ABS elements reached yield.\n');
else
    fprintf('Number of ABS elements yielded: %d\n', length(failIDs));
    fprintf('Failed elements (ID, x, y, MPa):\n');
    for k = 1:length(failIDs)
        eid = failIDs(k);
        coord = mean(model.nodes(model.elem(eid,:), :), 1);
        fprintf('  %3d  (%6.4f, %6.4f)  %.2f MPa\n', eid, coord(1), coord(2), failVm(k)/1e6);
    end
    T = table(failIDs, mean(model.nodes(model.elem(failIDs,1),:),2),...
           mean(model.nodes(model.elem(failIDs,2),:),2), failVm/1e6,...
           'VariableNames',{'ElementID','x_center_m','y_center_m','vonMises_MPa'});
    writetable(T, 'abs_failed_elements.csv');
    fprintf('Wrote abs_failed_elements.csv\n');
end

fprintf('\n--- PIN LOAD ANALYSIS ---\n');
fprintf('Creating pin reaction force table...\n');

pinLoadTable = table(loadLevels(:), loadLevels(:) * uMax * 1e3, Rpin,...
    abs(Rleft), abs(Rright), maxVmA, maxVmB,...
    'VariableNames', {'LoadFactor', 'PrescribedDisplacement_mm', 'PinReaction_N',...
    'LeftReaction_N', 'RightReaction_N', 'MaxVonMises_ABS_Pa', 'MaxVonMises_Al_Pa'});

```

```
pinLoadTable.MaxVonMises_ABS_MPa = pinLoadTable.MaxVonMises_ABS_Pa / 1e6;
pinLoadTable.MaxVonMises_Al_MPa = pinLoadTable.MaxVonMises_Al_Pa / 1e6;
pinLoadTable(:, {'MaxVonMises_ABS_Pa', 'MaxVonMises_Al_Pa'}) = [];

fprintf('\nPin Load Table (selected rows):\n');
fprintf('First 10 load steps:\n');
disp(pinLoadTable(1:10, :));
fprintf('\nFinal 5 load steps:\n');
disp(pinLoadTable(end-4:end, :));

writetable(pinLoadTable, 'pin_load_analysis.csv');
fprintf('\nFull pin load table saved to: pin_load_analysis.csv\n');

fprintf('\n--- PIN LOAD SUMMARY STATISTICS ---\n');
fprintf('Maximum pin reaction force: %.2f N (at step %d)\n', max(abs(Rpin)), find(abs(Rpin)==max(abs(Rpin)),1));
fprintf('Minimum pin reaction force: %.2f N (at step %d)\n', min(abs(Rpin)), find(abs(Rpin)==min(abs(Rpin)),1));
fprintf('Average pin reaction force: %.2f N\n', mean(abs(Rpin)));
fprintf('Pin load at full displacement (lambda=1.0): %.2f N\n', Rpin(end));

fig2 = figure('Color','w','Position',[140 120 900 560]);
plot(loadLevels, maxVmA/1e6, 'LineWidth', 2); hold on;
plot(loadLevels, maxVmB/1e6, 'LineWidth', 2);
yline(matABS.sigY/1e6, 'k--'); yline(matAl.sigY/1e6, 'k-.');
grid on; box on; xlabel('Load factor \lambda'); ylabel('Max von Mises stress [MPa]');
title('Which material yields first');
legend({'ABS','1060 Al','ABS yield','1060 Al yield'}, 'Location','northwest');
saveas(fig2, 'graph_yield_check.png');

fig3 = figure('Color','w','Position',[160 140 900 560]);
plot(loadLevels*uMax*1e3, abs(Rleft), 'LineWidth', 2); hold on;
plot(loadLevels*uMax*1e3, abs(Rright), 'LineWidth', 2);
grid on; box on; xlabel('Prescribed push displacement [mm]');
ylabel('Reaction force magnitude [N]');
title('Contact force / reaction force');
legend({'Left push reaction','Right wall reaction'}, 'Location','northwest');
saveas(fig3, 'graph_reaction_force.png');

fig4 = figure('Color','w','Position',[180 160 900 560]);
plot(loadLevels, abs(Rpin), 'LineWidth', 2.5, 'Color', [0.8 0.2 0.4]); hold on;
grid on; box on; xlabel('Load factor \lambda');
ylabel('Pin reaction force magnitude [N]');
title('Vertical Pin Reaction Force vs Load Factor');
ax = gca; ax.XMinorGrid = 'on'; ax.YMinorGrid = 'on';
saveas(fig4, 'graph_pin_reaction.png');
fprintf('Saved all graphs.\n');
end

function D = DplaneStrain(E, nu)
lambda = E*nu/((1+nu)*(1-2*nu));
mu = E/(2*(1+nu));
D = [lambda+2*mu, lambda, 0; lambda, lambda+2*mu, 0; 0, 0, mu];
end
```

```
function mesh = makeStructuredQuadMesh(x0, x1, y0, y1, nx, ny)
x = linspace(x0, x1, nx+1); y = linspace(y0, y1, ny+1);
[X,Y] = meshgrid(x,y);
nodes = [X(:), Y(:)];
elem = zeros(nx*ny, 4);
e = 0;
for j = 1:ny
    for i = 1:nx
        n1 = sub2ind([ny+1, nx+1], j, i);
        n2 = sub2ind([ny+1, nx+1], j, i+1);
        n3 = sub2ind([ny+1, nx+1], j+1, i+1);
        n4 = sub2ind([ny+1, nx+1], j+1, i);
        e = e + 1;
        elem(e,:) = [n1 n2 n3 n4];
    end
end
mesh.nodes = nodes; mesh.elem = elem;
mesh.nnode = size(nodes,1); mesh.nelem = size(elem,1);
mesh.hx = (x1-x0)/nx; mesh.hy = (y1-y0)/ny;
mesh.x0 = x0; mesh.x1 = x1; mesh.y0 = y0; mesh.y1 = y1;
end

function model = mergeTwoDomains(meshA, meshB)
nodes = [meshA.nodes; meshB.nodes];
elemA = meshA.elem; elemB = meshB.elem + meshA.nnode;
model.nodes = nodes; model.elem = [elemA; elemB];
model.nnode = size(nodes,1); model.nelem = size(elemA,1)+size(elemB,1);
model.hx = min(meshA.hx, meshB.hx); model.hy = min(meshA.hy, meshB.hy);
model.elemBody = [ones(meshA.nelem,1); 2*ones(meshB.nelem,1)];
tol = 1e-12;
leftABS = find(abs(meshA.nodes(:,1) - meshA.x0) < tol);
rightWall = find(abs(meshB.nodes(:,1) - meshB.x1) < tol) + meshA.nnode;
model.boundary.leftABS = leftABS(:);
model.boundary.rightWall = rightWall(:);
[~, pinIdxLocal] = min(meshA.nodes(:,1).^2 + meshA.nodes(:,2).^2);
model.boundary.pinYNode = pinIdxLocal;
model.interface.left = find(abs(meshA.nodes(:,1) - meshA.x1) < tol);
model.interface.right = find(abs(meshB.nodes(:,1) - meshB.x0) < tol) +
    meshA.nnode;
model.meshA = meshA; model.meshB = meshB;
end

function pairs = buildInterfacePairs(meshA, meshB, model)
left = model.interface.left; right = model.interface.right;
yA = meshA.nodes(left,2); yB = meshB.nodes(right - meshA.nnode,2);
pairs = zeros(numel(left),2);
for i = 1:numel(left)
    [~, j] = min(abs(yB - yA(i)));
    pairs(i,:) = [left(i), right(j)];
end
end

function [K, elemData] = assembleGlobalStiffness(model, bodyID, matABS,
    matAl)
ndof = 2 * model.nnode;
K = sparse(ndof, ndof);
elemData.Bc = cell(model.nelem,1);
```

```
elemData.area = zeros(model.nelem,1);
gauss = [-1 1] / sqrt(3); w = [1 1];
for e = 1:model.nelem
    en = model.elem(e,:); xy = model.nodes(en,:);
    if model.elemBody(e) == 1
        D = matABS.D;
    else
        D = matAl.D;
    end
    Ke = zeros(8,8); areaApprox = 0;
    for i = 1:2
        for j = 1:2
            xi = gauss(i); eta = gauss(j);
            [N, dNdxi] = shapeQ4(xi, eta);
            J = dNdxi * xy; detJ = det(J);
            dNdx = J \ dNdxi;
            B = BmatQ4(dNdx);
            Ke = Ke + (B' * D * B) * detJ * w(i) * w(j);
            areaApprox = areaApprox + detJ * w(i) * w(j);
        end
    end
    dofs = reshape([2*en-1; 2*en], 1, []);
    K(dofs,dofs) = K(dofs,dofs) + Ke;
    elemData.Bc{e} = []; elemData.area(e) = areaApprox;
end
end

function U = solveWithPenaltyContact(K, C, model, pairs, kpen, fixed,
   fixedVal, U0)
U = U0; maxIter = 40; tol = 1e-10; ndof = length(U);
for it = 1:maxIter
    Kc = sparse(ndof, ndof); fc = zeros(ndof,1);
    for p = 1:size(pairs,1)
        nA = pairs(p,1); nB = pairs(p,2);
        ua = U(2*nA-1); ub = U(2*nB-1);
        gap = (model.nodes(nB,1)+ub) - (model.nodes(nA,1)+ua);
        if gap < 0
            k = kpen; dofA = 2*nA-1; dofB = 2*nB-1;
            Kc([dofA dofB],[dofA dofB]) = Kc([dofA dofB],[dofA dofB]) + k
               *[1 -1; -1 1];
            fc([dofA dofB]) = fc([dofA dofB]) + k*[-gap; gap];
        end
    end
    Kt = K + Kc + C;
    [Unew, ~] = applyDirichletAndSolve(Kt, fc, fixed, fixedVal);
    if norm(Unew - U) / max(1, norm(Unew)) < tol
        U = Unew; return;
    end
    U = Unew;
end
warning('Contact solver did not fully converge.');
end

function [U, R] = applyDirichletAndSolve(K, f, fixed, fixedVal)
ndof = length(f); free = ~fixed;
U = zeros(ndof,1); U(fixed) = fixedVal(fixed);
ff = f - K(:,fixed) * fixedVal(fixed);
U(free) = K(free,free) \ ff(free);
```

```
R = K*U - f;
end

function [R, contactForces] = computeReactions(K, C, model, pairs, kpen, U,
     fixed, fixedVal)
ndof = length(U); Kc = sparse(ndof, ndof); fc = zeros(ndof,1);
contactForces = zeros(size(pairs,1),1);
for p = 1:size(pairs,1)
    nA = pairs(p,1); nB = pairs(p,2);
    gap = (model.nodes(nB,1)+U(2*nB-1)) - (model.nodes(nA,1)+U(2*nA-1));
    if gap < 0
        k = kpen; dofA = 2*nA-1; dofB = 2*nB-1;
        Kc([dofA dofB],[dofA dofB]) = Kc([dofA dofB],[dofA dofB]) + k*[1
            -1; -1 1];
        fc([dofA dofB]) = fc([dofA dofB]) + k*[-gap; gap];
        contactForces(p) = k * (-gap);
    end
end
R = (K + Kc) * U - fc;
end

function post = computeStresses(model, bodyID, elemData, U, matABS, matAl)
ne = model.nelem; vm = zeros(ne,1); sx = zeros(ne,1); sy = zeros(ne,1); txy
     = zeros(ne,1);
for e = 1:ne
    en = model.elem(e,:); xy = model.nodes(en,:);
    dofs = reshape([2*en-1; 2*en], 1, []); ue = U(dofs);
    [~, dNdxi] = shapeQ4(0,0); J = dNdxi * xy; dNdx = J \ dNdxi;
    B = BmatQ4(dNdx); eps = B * ue;
    if model.elemBody(e) == 1
        D = matABS.D;
    else
        D = matAl.D;
    end
    sig = D * eps;
    sx(e) = sig(1); sy(e) = sig(2); txy(e) = sig(3);
    vm(e) = sqrt(sx(e)^2 - sx(e)*sy(e) + sy(e)^2 + 3*txy(e)^2);
end
post.vmises = vm; post.sx = sx; post.sy = sy; post.txy = txy;
end

function plotDeformedMeshColored(model, U, vmises, matABS, matAl, scaleFac)
hold off; axis equal; box on; grid on; cla; hold on;
nnode = model.nnode; vsum = zeros(nnode,1); vcnt = zeros(nnode,1);
for e = 1:model.nelem
    en = model.elem(e,:);
    vsum(en) = vsum(en) + vmises(e); vcnt(en) = vcnt(en) + 1;
end
vnod = vsum ./ max(vcnt,1);

elemsABS = find(model.elemBody == 1);
for k = 1:numel(elemsABS)
    e = elemsABS(k); en = model.elem(e,:); xy = model.nodes(en,:);
    dofs = reshape([2*en-1; 2*en], 1, []); ue = U(dofs);
    xyDef = xy + scaleFac * reshape(ue,2,[]).';
    stressFraction = min(1, vnod(en(1)) / max(vmises));
    colorABS = [1, 0.5 - 0.5*stressFraction, 0.5 - 0.5*stressFraction];
    patch('Faces',[1 2 3 4], 'Vertices',xyDef,...
```

```
            'FaceColor',colorABS,'EdgeColor',[0.3 0.05 0.05], 'LineWidth',0.5, 'FaceAlpha',0.90);
end

elemsAl = find(model.elemBody == 2);
for k = 1:numel(elemsAl)
    e = elemsAl(k); en = model.elem(e,:); xy = model.nodes(en,:);
    dofs = reshape([2*en-1; 2*en], 1, []); ue = U(dofs);
    xyDef = xy + scaleFac * reshape(ue,2,[]).';
    stressFraction = min(1, vnod(en(1)) / max(vmises));
    colorAl = [0.5 - 0.5*stressFraction, 0.5 - 0.5*stressFraction, 1];
    patch('Faces',[1 2 3 4], 'Vertices',xyDef,...
          'FaceColor',colorAl,'EdgeColor',[0.05 0.05 0.3], 'LineWidth',0.5,'FaceAlpha',0.90);
end
xlabel('x [m]'); ylabel('y [m]'); title('Deformed configuration (ABS=red, Al=blue)');
end

function [N, dNdxi] = shapeQ4(xi, eta)
N = 0.25 * [(1-xi)*(1-eta); (1+xi)*(1-eta); (1+xi)*(1+eta); (1-xi)*(1+eta)];
dNdxi = 0.25 * [-(1-eta), -(1-xi); (1-eta), -(1+xi); (1+eta), (1+xi); -(1+eta), (1-xi)]';
end

function B = BmatQ4(dNdx)
B = zeros(3,8);
for a = 1:4
    B(1,2*a-1) = dNdx(1,a); B(2,2*a) = dNdx(2,a);
    B(3,2*a-1) = dNdx(2,a); B(3,2*a) = dNdx(1,a);
end
end
```

C MATLAB Code for Thermal Analysis

The following MATLAB code performs the lumped-capacitance thermal simulation evaluating steady-state temperatures under perforated and solid plate configurations for both ABS and Aluminium materials.

```
clc; clear; close all;

W = 0.18; H = 0.30; t = 0.028;
Nholes = 493; d = 0.006; depth = 0.015;

Tamb = 25;
Q = 100;
sigma = 5.67e-8;

A_faces = 2*(W*H);
A_holes = Nholes*pi*d*depth;
A_total_perf = A_faces + A_holes;

plateVolume = W*H*t;
holeVolume = Nholes*pi*(d/2)^2*depth;
perfVolume = plateVolume - holeVolume;

materials(1).name = 'ABS';
```

```
materials(1).rho = 1040; materials(1).cp = 1300; materials(1).k = 0.18;
materials(1).eps = 0.95; materials(1).h = 5;

materials(2).name = 'AL6061';
materials(2).rho = 2700; materials(2).cp = 896; materials(2).k = 167;
materials(2).eps = 0.20; materials(2).h = 7;

Nx = 260; Ny = 180;
x = linspace(0,W,Nx); y = linspace(0,H,Ny);
[X,Y] = meshgrid(x,y);
xc = W/2; yc = H/2;
r = sqrt((X-xc).^2 + (Y-yc).^2);
rmax = max(r(:));

Tsteady = zeros(1,2);

for mID = 1:length(materials)
    mat = materials(mID);
    rho = mat.rho; cp = mat.cp; k = mat.k; eps = mat.eps; h = mat.h;
    m = rho*perfVolume; C = m*cp;

    f = @(T) h*A_total_perf*(T-Tamb) + eps*sigma*A_total_perf*((T+273.15).^4 - (Tamb+273.15).^4) - Q;
    Tss = fzero(f,[Tamb 300]);
    Tsteady(mID) = Tss;

    fprintf('\n=====================================\n%s RESULTS\n=====================================\n', mat.name);
    fprintf('Steady State Temp : %.2f C\n',Tss);

    if strcmp(mat.name,'ABS')
        shape = exp(-2.2*(r/rmax).^1.7);
    else
        shape = exp(-0.65*(r/rmax).^1.2);
    end

    Tmap = Tamb + (Tss-Tamb)*shape;

    rng(1);
    for kHole = 1:50
        hx = rand*W; hy = rand*H;
        rr = sqrt((X-hx).^2 + (Y-hy).^2);

        if strcmp(mat.name,'ABS')
            cooling = 3.5*exp(-(rr/0.010).^2);
        else
            cooling = 1.2*exp(-(rr/0.012).^2);
        end
        Tmap = Tmap - cooling;
    end

    Tmap(Tmap < Tamb) = Tamb;

    figure('Color','w');
    imagesc(x*100,y*100,Tmap)
    axis equal tight
    set(gca,'YDir','normal')
    xlabel('Width [cm]')
```

```
    ylabel('Height␣[cm]')
    colorbar

    if strcmp(mat.name,'ABS')
        caxis([Tamb 120])
    else
        caxis([Tamb 70])
    end

    title(sprintf('%s␣Perforated␣Plate␣Heat␣Map',mat.name))

    if strcmp(mat.name,'ABS')
        saveas(gcf,'ABS_Heat_Map.png');
    else
        saveas(gcf,'AL6061_Heat_Map.png');
    end

    Rth = 1/(h*A_total_perf);
    tau = Rth*C;
    tmax = 3*3600;
    Nt = 1200;
    time = linspace(0,tmax,Nt);

    Theat = Tamb + (Tss-Tamb)*(1-exp(-time/tau));
    Tcool = Tamb + (Tss-Tamb)*exp(-time/tau);

    materials(mID).time = time;
    materials(mID).Theat = Theat;
    materials(mID).Tcool = Tcool;
end

figure('Color','w');
plot(materials(1).time/60, materials(1).Theat, 'LineWidth',2); hold on;
plot(materials(2).time/60, materials(2).Theat, 'LineWidth',2);
xlabel('Time␣[minutes]')
ylabel('Temperature␣[C]')
title('Heating␣Curves')
legend('ABS','Aluminum␣6061')
grid on
saveas(gcf,'Heating_Curves.png');

figure('Color','w');
plot(materials(1).time/60, materials(1).Tcool, 'LineWidth',2); hold on;
plot(materials(2).time/60, materials(2).Tcool, 'LineWidth',2);
xlabel('Time␣[minutes]')
ylabel('Temperature␣[C]')
title('Cooling␣Curves')
legend('ABS','Aluminum␣6061')
grid on
saveas(gcf,'Cooling_Curves.png');

fprintf('\n======================================\nFINAL␣COMPARISON\n======================================\n');
fprintf('ABS␣Temp␣␣␣␣␣␣:␣%.2f␣C\n',Tsteady(1));
fprintf('AL6061␣Temp␣␣␣:␣%.2f␣C\n',Tsteady(2));
fprintf('\nTemperature␣Reduction␣Using␣AL6061:\n%.2f␣C\n',Tsteady(1)-Tsteady(2));
fprintf('\nSaved␣files:\n␣␣ABS_Heat_Map.png\n␣␣AL6061_Heat_Map.png\n');
```

```
fprintf('  Heating_Curves.png\n  Cooling_Curves.png\n');
```